Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Pires Dairy Farms #2 (Formerly Jones Farms)

Physical address of dairy:

Number and Street	City	County	Zip Code
1511 Van Clief RD	Stevinson	Merced	95374

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 05/01/1994

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X055-X235-X002-XXXX

B. OPERATORS

Pires, Eldred & Amanda

Operator name: Pires, Eldred & Amanda

Telephone no.: (209) 652-3530 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
1597 Kniebes RD	Gustine	CA	95322

This operator is responsible for paying permit fees.

C. OWNERS

Jones, Rick & Julie

Legal owner name: Jones, Rick & Julie

Telephone no.: (209) 634-4754 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
19133 W Sixth AVE	Stevinson	CA	95324

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	672	0	360	200	0	0
Maximum number	690	0	360	200	0	0
Average number	672	0	360	200	0	0
Avg live weight (lbs)	1,400	0	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 84 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 24,173.11 *tons per reporting period*

Total nitrogen from manure: 292,955.63 *lbs per reporting period*

After ammonia losses (30% loss applied): 205,068.94 *lbs per reporting period*

Total phosphorus from manure: 49,802.72 *lbs per reporting period*

Total potassium from manure: 134,875.26 *lbs per reporting period*

Total salt from manure: 316,411.20 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 19,993,456 *gallons*

Total nitrogen generated: 74,203.78 *lbs*

Total phosphorus generated: 11,016.42 *lbs*

Total potassium generated: 110,663.87 *lbs*

Total salt generated: 702,916.55 *lbs*

	19,993,456	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	19,993,456	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/11/2023	Solid commercial fertilizer *UN32*	40.16 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	25,676.70	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	25,676.70	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
F1	19	19	1	none	X055-X235-X018-XXXX
F10	13	13	2	process wastewater	X055-X236-X012-XXXX
F11	29	29	2	process wastewater	X055-X236-X013-XXXX X055-X236-X014-XXXX
F12	27	27	1	none	X055-X330-X004-XXXX X055-X330-X005-XXXX X055-X330-X006-XXXX X055-X330-X031-XXXX
F13	17	17	1	none	X055-X330-X025-XXXX
F14	36	36	1	none	X055-X330-X024-XXXX
F2	22	22	2	process wastewater	X055-X235-X009-XXXX
F3	23	23	2	process wastewater	X055-X235-X002-XXXX
F4	32	32	2	process wastewater	X055-X235-X005-XXXX X055-X235-X006-XXXX
F5	21	21	2	process wastewater	X055-X235-X004-XXXX
F6	6	6	1	manure	X055-X235-X003-XXXX
F7	18	18	2	process wastewater	X055-X238-X034-XXXX X055-X238-X054-XXXX
F8	57	57	2	manure	X055-X238-X037-XXXX X055-X238-X038-XXXX X055-X238-X039-XXXX X055-X238-X040-XXXX X055-X238-X041-XXXX X055-X238-X051-XXXX
F9	18	18	2	both	X055-X233-X006-XXXX
R1	22	22	1	none	X055-X234-X010-XXXX
R2	12	12	1	manure	X055-X237-X002-XXXX
R3	26	26	2	process wastewater	X055-X238-X025-XXXX
R4	22	22	2	manure	X055-X238-X005-XXXX X055-X238-X052-XXXX X055-X238-X053-XXXX
R5/R6	9	9	2	both	X055-X236-X001-XXXX X055-X236-X002-XXXX
R8	18	18	2	manure	X055-X350-X013-XXXX

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
R9	13	13	2	manure	X055-X340-X019-XXXX X055-X340-X020-XXXX
Totals for areas that were used for application	339	339	30		
Totals for areas that were not used for application	121	121	5		
Land application area totals	460	460	35		

B. CROPS AND HARVESTS

F1

Field name: F1

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 19 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	159.60 *ton*	As-is		9.7	30,300.00	3,000.00	23,900.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.40	509.04	50.40	401.52	1,774.94

F10

Field name: F10

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	250.92 *ton*	As-is		58.0	5,900.00	1,500.00	4,400.00		9.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.30	227.76	57.90	169.85	1,517.56

F10

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	352.17 *ton*	As-is		68.8	3,500.00	500.00	4,100.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.09	189.63	27.09	222.14	2,096.12

F11

Field name: F11

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	518.38 *ton*	As-is		57.9	5,500.00	1,700.00	4,600.00		8.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.88	196.63	60.78	164.45	1,327.49

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	741.31 *ton*	As-is		68.4	3,300.00	500.00	3,700.00		4.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.56	168.71	25.56	189.16	694.68

F12

Field name: F12

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 27 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	248.40 *ton*	As-is		10.2	31,300.00	3,400.00	24,400.00		14.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.20	575.92	62.56	448.96	2,412.39

F13

Field name: F13

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 17 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	142.80 *ton*	As-is		8.8	33,500.00	3,100.00	29,700.00		13.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.40	562.80	52.08	498.96	2,053.09

F14

Field name: F14

F14

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 36 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	280.80 *ton*	As-is		8.8	33,500.00	3,100.00	29,700.00		13.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.80	522.60	48.36	463.32	1,906.44

F2

Field name: F2

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	394.00 *ton*	As-is		57.5	5,300.00	1,900.00	4,700.00		9.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.91	189.84	68.05	168.35	1,467.47

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	676.19 *ton*	As-is		72.2	3,500.00	500.00	4,600.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.74	215.15	30.74	282.77	1,144.97

F3

Field name: F3

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	494.05 *ton*	As-is		63.1	4,700.00	1,400.00	4,200.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	21.48	201.92	60.15	180.44	1,632.81

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	670.68 *ton*	As-is		71.2	3,600.00	500.00	4,200.00		4.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.16	209.95	29.16	244.94	688.64

F4

Field name: F4

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	593.61 *ton*	As-is		61.0	5,100.00	1,500.00	4,300.00		9.14

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.55	189.21	55.65	159.53	1,322.49

F4

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	1,000.64 *ton*	As-is		74.4	3,000.00	400.00	3,600.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.27	187.62	25.02	225.14	784.50

F5

Field name: F5

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	427.20 *ton*	As-is		69.2	4,200.00	1,100.00	5,200.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.34	170.88	44.75	211.57	1,416.03

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	656.67 *ton*	As-is		71.2	3,600.00	500.00	4,200.00		4.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.27	225.14	31.27	262.67	738.47

F6

Field name: F6

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 6 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	27.00 *ton*	As-is		9.7	32,800.00	3,600.00	22,000.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	32.00	16.00
Total actual harvest content	4.50	295.20	32.40	198.00	1,056.51

F7

Field name: F7

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	370.44 *ton*	As-is		60.1	4,700.00	1,400.00	6,300.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.58	193.45	57.62	259.31	1,970.74

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	460.69 *ton*	As-is		69.4	3,400.00	500.00	4,200.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.59	174.04	25.59	214.99	845.83

F8

Field name: F8

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	798.21 *ton*	As-is		67.7	6,300.00	1,200.00	6,900.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	176.45	33.61	193.25	1,013.19

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	1,698.53 *ton*	As-is		70.2	3,400.00	500.00	4,000.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.80	202.63	29.80	238.39	941.28

F9

Field name: F9

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	206.38 *ton*	As-is		58.0	5,900.00	1,500.00	4,400.00		9.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.47	135.29	34.40	100.90	901.47

F9

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	545.74 *ton*	As-is		68.8	3,500.00	500.00	4,100.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.32	212.23	30.32	248.61	2,345.95

R1

Field name: R1

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 22 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	193.60 *ton*	As-is		8.6	34,500.00	3,600.00	21,800.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.80	607.20	63.36	383.68	1,946.45

R2

Field name: R2

R2

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 12 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	61.20 *ton*	As-is		8.6	34,500.00	3,600.00	21,800.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	5.10	351.90	36.72	222.36	1,128.06

R3

Field name: R3

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/06/2023	415.45 *ton*	As-is		63.3	4,700.00	1,000.00	5,600.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.98	150.20	31.96	178.96	1,337.05

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	833.15 *ton*	As-is		71.6	3,400.00	400.00	4,200.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.04	217.90	25.64	269.17	1,164.87

R4

Field name: R4

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/06/2023	322.68 *ton*	As-is		63.3	4,700.00	1,000.00	5,600.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.67	137.87	29.33	164.27	1,227.30

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	572.25 *ton*	As-is		73.4	3,300.00	400.00	4,100.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.01	171.68	20.81	213.29	816.44

R5/R6

Field name: R5/R6

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	127.00 *ton*	As-is		63.3	4,700.00	1,000.00	5,600.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.11	132.64	28.22	158.04	1,180.76

R5/R6

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	228.45 *ton*	As-is		67.0	3,600.00	400.00	3,900.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.38	182.76	20.31	197.99	938.17

R8

Field name: R8

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/06/2023	289.00 *ton*	As-is		65.5	5,200.00	1,100.00	5,700.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.06	166.98	35.32	183.03	1,362.64

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	430.25 *ton*	As-is		68.2	3,300.00	600.00	4,100.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.90	157.76	28.68	196.00	699.30

R9

Field name: R9

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/06/2023	210.00 *ton*	As-is		70.1	5,500.00	900.00	5,400.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.15	177.69	29.08	174.46	1,101.24

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	358.10 *ton*	As-is		68.2	3,300.00	600.00	4,100.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.55	181.80	33.06	225.88	805.89

NUTRIENT BUDGET

A. LAND APPLICATIONS

F1 - 01/01/2023: Alfalfa, hay

Field name: F1

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	46.25	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	33.53	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	33.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	23.12	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	24.28	2,126,250.00 *gal*
Application event totals		0.00	0.00	0.00	24.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	28.91	2,531,250.00 *gal*
Application event totals		0.00	0.00	0.00	28.91	

F1 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.66	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	19.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	13.87	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	23.12	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.12	

F10 - 11/03/2022: Oats, silage-soft dough

Field name: F10

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.37	14.03	127.12	1,164.13	292,500.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	563.29	900,000.00 *gal*
Irrigation Well	Ground water	5.78	0.00	0.00	896.64	1,440,000.00 *gal*
Application event totals		95.15	14.03	127.12	2,624.05	

F10 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.00	8.63	78.22	716.39	180,000.00 *gal*
Application event totals		55.00	8.63	78.22	716.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.54	14.04	84.23	410.77	405,000.00 *gal*
Application event totals		46.54	14.04	84.23	410.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.78	11.70	70.19	342.31	337,500.00 *gal*
Irrigation Well	Ground water	1.89	0.00	0.00	541.51	819,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	492.88	787,500.00 *gal*
Application event totals		40.67	11.70	70.19	1,376.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	40.56	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.56	

F10 - 05/31/2023: Corn, silage

Field name: F10

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F10 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.07	6.39	98.07	525.74	157,500.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	868.40	1,387,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	30.42	1,822,500.00 *gal*
Application event totals		52.07	6.39	98.07	1,424.55	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.78	0.00	0.00	896.64	1,440,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	32.11	1,923,750.00 *gal*
Application event totals		5.78	0.00	0.00	928.75	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	20.28	1,215,000.00 *gal*
Application event totals		80.00	0.00	0.00	20.28	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.19	4.56	70.05	375.53	112,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	18.59	1,113,750.00 *gal*
Application event totals		37.19	4.56	70.05	394.11	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.51	7.30	112.08	600.84	180,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	539.82	862,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	13.52	810,000.00 *gal*
Application event totals		59.51	7.30	112.08	1,154.18	

F10 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.75	3.65	56.04	300.42	90,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	21.97	1,316,250.00 *gal*
Application event totals		29.75	3.65	56.04	322.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	13.52	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.52	

F11 - 11/02/2022: Oats, silage-soft dough

Field name: F11

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.37	14.03	127.12	1,164.13	652,500.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	389.28	1,387,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	30.30	4,050,000.00 *gal*
Application event totals		89.37	14.03	127.12	1,583.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.81	8.39	50.35	245.52	540,000.00 *gal*
Application event totals		27.81	8.39	50.35	245.52	

F11 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.54	5.59	33.56	163.68	360,000.00 *gal*
Application event totals		18.54	5.59	33.56	163.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.07	4.55	27.27	132.99	292,500.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	336.68	1,200,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	25.76	3,442,500.00 *gal*
Application event totals		15.07	4.55	27.27	495.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.35	6.55	100.49	538.68	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	272.71	36,450,000.00 *gal*
Application event totals		53.35	6.55	100.49	811.39	

F11 - 05/29/2023: Corn, silage

Field name: F11

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.68	5.73	87.92	471.35	315,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	473.45	1,687,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	36.36	4,860,000.00 *gal*
Application event totals		46.68	5.73	87.92	981.16	

F11 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	31.82	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	31.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Irrigation Well	Ground water	0.00	0.00	0.00	273.55	975,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	21.97	2,936,250.00 *gal*
Application event totals		80.00	0.00	0.00	295.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.34	4.09	62.80	336.68	225,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	15.15	2,025,000.00 *gal*
Application event totals		33.34	4.09	62.80	351.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.35	6.55	100.49	538.68	360,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	305.11	1,087,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	27.27	3,645,000.00 *gal*
Application event totals		53.35	6.55	100.49	871.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	13.64	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	13.64	

F11 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.68	3.27	50.24	269.34	180,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	210.42	750,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	18.18	2,430,000.00 *gal*
Application event totals		26.68	3.27	50.24	497.95	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	15.91	2,126,250.00 *gal*
Application event totals		0.00	0.00	0.00	15.91	

F12 - 01/01/2023: Alfalfa, hay

Field name: F12

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	30.10	3,746,250.00 *gal*
Application event totals		0.00	0.00	0.00	30.10	

05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	22.78	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.78	

F12 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	17.90	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	17.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	33.36	4,151,250.00 *gal*
Application event totals		0.00	0.00	0.00	33.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.53	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	13.83	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	13.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	23.60	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	23.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.53	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

F12 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.29	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

F13 - 01/01/2023: Alfalfa, hay

Field name: F13

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	37.48	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	37.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	23.26	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	23.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	31.01	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	16.80	1,316,250.00 *gal*
Application event totals		0.00	0.00	0.00	16.80	

F13 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	21.97	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	31.01	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	20.68	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	14.21	1,113,750.00 *gal*
Application event totals		0.00	0.00	0.00	14.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	21.97	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

F14 - 01/01/2023: Alfalfa, hay

Field name: F14

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F14 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.78	5,771,250.00 *gal*
Application event totals		0.00	0.00	0.00	34.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	25.63	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	25.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	21.97	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.46	4,556,250.00 *gal*
Application event totals		0.00	0.00	0.00	27.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.53	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	18.31	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	18.31	

F14 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	25.02	4,151,250.00 *gal*
Application event totals		0.00	0.00	0.00	25.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	21.97	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.53	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

F2 - 10/16/2022: Oats, silage-soft dough

Field name: F2

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.00	10.20	92.45	846.64	360,000.00 *gal*
Irrigation Well	Ground water	4.55	0.00	0.00	706.44	1,920,000.00 *gal*
Irrigation Well	Ground water	1.70	0.00	0.00	487.59	1,248,000.00 *gal*
Application event totals		71.25	10.20	92.45	2,040.67	

F2 - 10/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.81	8.29	75.11	687.89	292,500.00 *gal*
Application event totals		52.81	8.29	75.11	687.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.50	8.30	49.77	242.73	405,000.00 *gal*
Application event totals		27.50	8.30	49.77	242.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.33	5.53	33.18	161.82	270,000.00 *gal*
Irrigation Well	Ground water	3.84	0.00	0.00	596.06	1,620,000.00 *gal*
Irrigation Well	Ground water	1.43	0.00	0.00	411.40	1,053,000.00 *gal*
Application event totals		23.61	5.53	33.18	1,169.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.95	5.39	82.79	443.80	225,000.00 *gal*
Irrigation Well	Ground water	0.85	0.00	0.00	243.80	624,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	31.95	3,240,000.00 *gal*
Application event totals		44.80	5.39	82.79	719.55	

F2 - 05/31/2023: Corn, silage

Field name: F2

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F2 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.14	7.01	107.62	576.94	292,500.00 *gal*
Irrigation Well	Ground water	1.65	0.00	0.00	472.35	1,209,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	31.95	3,240,000.00 *gal*
Application event totals		58.79	7.01	107.62	1,081.25	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	1.27	0.00	0.00	365.69	936,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	41.94	4,252,500.00 *gal*
Application event totals		1.27	0.00	0.00	407.63	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	33.95	3,442,500.00 *gal*
Application event totals		80.00	0.00	0.00	33.95	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.35	5.93	91.06	488.18	247,500.00 *gal*
Irrigation Well	Ground water	1.12	0.00	0.00	319.98	819,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	17.97	1,822,500.00 *gal*
Application event totals		49.46	5.93	91.06	826.14	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.16	4.32	66.23	355.04	180,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	19.97	2,025,000.00 *gal*
Application event totals		35.16	4.32	66.23	375.01	

F2 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	14.98	1,518,750.00 *gal*
Application event totals		0.00	0.00	0.00	14.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.77	3.78	57.95	310.66	157,500.00 *gal*
Irrigation Well	Ground water	0.53	0.00	0.00	152.37	390,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	17.97	1,822,500.00 *gal*
Application event totals		31.30	3.78	57.95	481.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	11.98	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.98	

F3 - 11/04/2022: Oats, silage-soft dough

Field name: F3

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.92	16.47	149.22	1,366.58	607,500.00 *gal*
Irrigation Well	Ground water	4.63	0.00	0.00	717.96	2,040,000.00 *gal*
Irrigation Well	Ground water	1.73	0.00	0.00	495.54	1,326,000.00 *gal*
Application event totals		111.27	16.47	149.22	2,580.09	

F3 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.84	7.49	44.97	219.27	382,500.00 *gal*
Application event totals		24.84	7.49	44.97	219.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.07	10.58	63.48	309.56	540,000.00 *gal*
Application event totals		35.07	10.58	63.48	309.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.27	0.00	0.00	506.80	1,440,000.00 *gal*
Irrigation Well	Ground water	1.22	0.00	0.00	349.79	936,000.00 *gal*
Application event totals		4.48	0.00	0.00	856.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.25	5.68	87.11	466.96	247,500.00 *gal*
Irrigation Well	Ground water	1.41	0.00	0.00	403.61	1,080,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	34.39	3,645,000.00 *gal*
Application event totals		47.65	5.68	87.11	904.95	

F3 - 06/01/2023: Corn, silage

Field name: F3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.27	8.26	126.70	679.21	360,000.00 *gal*
Irrigation Well	Ground water	2.29	0.00	0.00	655.86	1,755,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	30.56	3,240,000.00 *gal*
Application event totals		69.55	8.26	126.70	1,365.64	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.70	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	27.70	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	25.79	2,733,750.00 *gal*
Application event totals		80.00	0.00	0.00	25.79	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.04	5.16	79.19	424.51	225,000.00 *gal*
Irrigation Well	Ground water	1.32	0.00	0.00	378.94	1,014,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	23.88	2,531,250.00 *gal*
Application event totals		43.36	5.16	79.19	827.33	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.43	3.61	55.43	297.15	157,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	28.65	3,037,500.00 *gal*
Application event totals		29.43	3.61	55.43	325.81	

F3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.84	4.64	71.27	382.06	202,500.00 *gal*
Irrigation Well	Ground water	1.17	0.00	0.00	335.22	897,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	22.92	2,430,000.00 *gal*
Application event totals		39.01	4.64	71.27	740.20	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	16.24	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	16.24	

F4 - 11/02/2022: Oats, silage-soft dough

Field name: F4

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.17	17.29	156.69	1,434.95	887,500.00 *gal*
Irrigation Well	Ground water	3.62	0.00	0.00	561.57	2,220,000.00 *gal*
Irrigation Well	Ground water	1.35	0.00	0.00	387.60	1,443,000.00 *gal*
Application event totals		115.14	17.29	156.69	2,384.11	

12/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.72	5.92	53.65	491.32	303,875.00 *gal*
Application event totals		37.72	5.92	53.65	491.32	

F4 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.48	9.19	55.16	269.01	652,889.00 *gal*
Application event totals		30.48	9.19	55.16	269.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.25	10.63	63.80	311.14	755,141.00 *gal*
Irrigation Well	Ground water	4.30	0.00	0.00	667.81	2,640,000.00 *gal*
Application event totals		39.55	10.63	63.80	978.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.13	0.00	0.00	485.68	1,920,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	43.94	6,480,000.00 *gal*
Application event totals		3.13	0.00	0.00	529.62	

F4 - 06/01/2023: Corn, silage

Field name: F4

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	163.14	9.97	66.76	534.08	160,000.00 *gal*
Application event totals		163.14	9.97	66.76	534.08	

F4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.86	0.00	0.00	599.51	2,370,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	41.19	6,075,000.00 *gal*
Application event totals		3.86	0.00	0.00	640.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	32.95	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Irrigation Well	Ground water	2.84	0.00	0.00	440.15	1,740,000.00 *gal*
Application event totals		82.84	0.00	0.00	440.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	23.34	3,442,500.00 *gal*
Application event totals		0.00	0.00	0.00	23.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.28	0.00	0.00	508.45	2,010,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	27.46	4,050,000.00 *gal*
Application event totals		3.28	0.00	0.00	535.91	

F4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.91	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	19.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	24.71	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.71	

F5 - 11/02/2022: Oats, silage-soft dough

Field name: F5

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	755.52	1,950,000.00 *gal*
Application event totals		0.00	0.00	0.00	755.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.03	12.56	113.83	1,042.42	423,100.00 *gal*
Application event totals		80.03	12.56	113.83	1,042.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.86	12.93	77.58	378.34	602,578.00 *gal*
Application event totals		42.86	12.93	77.58	378.34	

F5 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.89	9.02	54.10	263.82	420,189.00 *gal*
Application event totals		29.89	9.02	54.10	263.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	682.87	1,762,500.00 *gal*
Application event totals		0.00	0.00	0.00	682.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.19	5.67	86.99	466.35	225,684.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	581.17	1,500,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	41.84	4,050,000.00 *gal*
Application event totals		46.19	5.67	86.99	1,089.36	

F5 - 05/30/2023: Corn, silage

Field name: F5

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	163.14	9.97	66.76	534.08	105,000.00 *gal*
Application event totals		163.14	9.97	66.76	534.08	

F5 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	479.46	1,237,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	42.89	4,151,250.00 *gal*
Application event totals		0.00	0.00	0.00	522.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	30.34	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	30.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Irrigation Well	Ground water	0.00	0.00	0.00	348.70	900,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	33.48	3,240,000.00 *gal*
Application event totals		80.00	0.00	0.00	382.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	290.58	750,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.88	1,923,750.00 *gal*
Application event totals		0.00	0.00	0.00	19.88	

F5 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	20.92	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	247.00	637,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.57	3,442,500.00 *gal*
Application event totals		0.00	0.00	0.00	282.56	

F6 - 01/01/2023: Alfalfa, hay

Field name: F6

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.96	61.65	172.03	6,473.54	66.00 *ton*
Application event totals		184.96	61.65	172.03	6,473.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	152.56	112,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.29	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	181.85	

F6 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	508.52	375,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	14.65	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	523.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.29	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	432.24	318,750.00 *gal*
Application event totals		0.00	0.00	0.00	432.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	14.65	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	355.97	262,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	21.97	607,500.00 *gal*
Application event totals		0.00	0.00	0.00	377.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	36.61	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	36.61	

F6 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	355.97	262,500.00 *gal*
Application event totals		0.00	0.00	0.00	355.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	355.97	262,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	18.31	506,250.00 *gal*
Application event totals		0.00	0.00	0.00	374.27	

F7 - 10/16/2022: Oats, silage-soft dough

Field name: F7

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.41	13.25	120.05	1,099.45	382,500.00 *gal*
Irrigation Well	Ground water	2.55	0.00	0.00	342.15	1,476,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	694.98	1,537,500.00 *gal*
Application event totals		86.96	13.25	120.05	2,136.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.54	6.20	37.18	181.30	247,500.00 *gal*
Application event totals		20.54	6.20	37.18	181.30	

F7 - 10/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.41	6.76	40.56	197.78	270,000.00 *gal*
Application event totals		22.41	6.76	40.56	197.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.24	0.00	0.00	300.42	1,296,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	976.37	2,160,000.00 *gal*
Application event totals		2.24	0.00	0.00	1,276.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.88	9.01	54.08	263.70	360,000.00 *gal*
Application event totals		29.88	9.01	54.08	263.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.72	6.59	101.18	542.43	225,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.29	2,430,000.00 *gal*
Application event totals		53.72	6.59	101.18	571.72	

F7 - 05/30/2023: Corn, silage

Field name: F7

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F7 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.47	7.91	121.42	650.91	270,000.00 *gal*
Irrigation Well	Ground water	2.30	0.00	0.00	308.77	1,332,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	43.94	3,645,000.00 *gal*
Application event totals		66.76	7.91	121.42	1,003.61	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	1.74	0.00	0.00	233.66	1,008,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	30.51	2,531,250.00 *gal*
Application event totals		1.74	0.00	0.00	264.17	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	19.53	1,620,000.00 *gal*
Application event totals		80.00	0.00	0.00	19.53	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.35	5.93	91.06	488.18	202,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	25.63	2,126,250.00 *gal*
Application event totals		48.35	5.93	91.06	513.81	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	1.55	0.00	0.00	208.63	900,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	12.20	1,012,500.00 *gal*
Application event totals		1.55	0.00	0.00	220.83	

F7 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.23	3.96	60.71	325.46	135,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	19.53	1,620,000.00 *gal*
Application event totals		32.23	3.96	60.71	344.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.49	2.64	40.47	216.97	90,000.00 *gal*
Irrigation Well	Ground water	1.30	0.00	0.00	175.25	756,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	17.09	1,417,500.00 *gal*
Application event totals		22.79	2.64	40.47	409.30	

F8 - 10/17/2022: Oats, silage-soft dough

Field name: F8

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.96	61.65	172.03	6,473.54	627.00 *ton*
Application event totals		184.96	61.65	172.03	6,473.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	942.11	6,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	942.11	

F8 - 10/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	779.38	5,460,000.00 *gal*
Application event totals		0.00	0.00	0.00	779.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	719.43	5,040,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	43.17	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	762.59	

F8 - 05/29/2023: Corn, silage

Field name: F8

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	158.40	43.56	182.16	15,166.80	627.00 *ton*
Application event totals		158.40	43.56	182.16	15,166.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	830.77	5,820,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	24.67	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	855.43	

F8 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.27	5,062,500.00 *gal*
Application event totals		0.00	0.00	0.00	19.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Irrigation Well	Ground water	0.00	0.00	0.00	950.67	6,660,000.00 *gal*
Application event totals		80.00	0.00	0.00	950.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	659.47	4,620,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	20.04	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	679.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	18.50	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	13.10	3,442,500.00 *gal*
Application event totals		0.00	0.00	0.00	13.10	

F8 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	719.43	5,040,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	16.19	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	735.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	12.33	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.33	

F9 - 11/03/2022: Oats, silage-soft dough

Field name: F9

Crop: Oats, silage-soft dough Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.96	61.65	172.03	6,473.54	198.00 *ton*
Application event totals		184.96	61.65	172.03	6,473.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	51.26	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	51.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	39.05	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

F9 - 05/31/2023: Corn, silage

Field name: F9

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	163.14	9.97	66.76	534.08	90,000.00 *gal*
Application event totals		163.14	9.97	66.76	534.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	41.50	3,442,500.00 *gal*
Application event totals		0.00	0.00	0.00	41.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	32.95	2,733,750.00 *gal*
Application event totals		0.00	0.00	0.00	32.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	20.75	1,721,250.00 *gal*
Application event totals		80.00	0.00	0.00	20.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	24.41	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.41	

F9 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	32.95	2,733,750.00 *gal*
Application event totals		0.00	0.00	0.00	32.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.53	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	20.75	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	20.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	24.41	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.41	

R1 - 01/01/2023: Alfalfa, hay

Field name: R1

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	39.94	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.94	

R1 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	28.96	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	28.96	

05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.97	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.97	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.96	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.96	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	23.97	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.97	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	21.97	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	16.98	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	16.98	

R1 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.97	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	15.98	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.98	

R2 - 01/01/2023: Alfalfa, hay

Field name: R2

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.96	61.65	172.03	6,473.54	132.00 *ton*
Application event totals		184.96	61.65	172.03	6,473.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	40.28	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	40.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	23.80	1,316,250.00 *gal*
Application event totals		0.00	0.00	0.00	23.80	

R2 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	14.65	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.29	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	16.48	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	16.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	12.81	708,750.00 *gal*
Application event totals		0.00	0.00	0.00	12.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	21.97	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	14.65	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.65	

R2 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	10.98	607,500.00 *gal*
Application event totals		0.00	0.00	0.00	10.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.29	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

R3 - 10/16/2022: Oats, silage-soft dough

Field name: R3

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	1.76	0.00	0.00	236.87	1,476,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	33.80	4,050,000.00 *gal*
Application event totals		1.76	0.00	0.00	270.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.50	12.95	117.34	1,074.58	540,000.00 *gal*
Application event totals		82.50	12.95	117.34	1,074.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.56	7.41	44.46	216.79	427,500.00 *gal*
Application event totals		24.56	7.41	44.46	216.79	

R3 - 10/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.98	6.63	39.78	193.97	382,500.00 *gal*
Irrigation Well	Ground water	1.50	0.00	0.00	202.21	1,260,000.00 *gal*
Application event totals		23.48	6.63	39.78	396.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.38	9.13	140.10	751.05	450,000.00 *gal*
Irrigation Well	Ground water	1.46	0.00	0.00	196.43	1,224,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	30.42	3,645,000.00 *gal*
Application event totals		75.84	9.13	140.10	977.90	

R3 - 06/01/2023: Corn, silage

Field name: R3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.10	9.58	147.10	788.60	472,500.00 *gal*
Irrigation Well	Ground water	1.55	0.00	0.00	207.98	1,296,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	33.80	4,050,000.00 *gal*
Application event totals		79.65	9.58	147.10	1,030.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	26.19	3,138,750.00 *gal*
Application event totals		0.00	0.00	0.00	26.19	

R3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	20.28	2,430,000.00 *gal*
Application event totals		80.00	0.00	0.00	20.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.79	6.85	105.07	563.29	337,500.00 *gal*
Irrigation Well	Ground water	0.69	0.00	0.00	92.44	576,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.49	4,252,500.00 *gal*
Application event totals		56.47	6.85	105.07	691.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.60	0.00	0.00	80.88	504,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	32.11	3,847,500.00 *gal*
Application event totals		0.60	0.00	0.00	112.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.75	3.65	56.04	300.42	180,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	27.04	3,240,000.00 *gal*
Application event totals		29.75	3.65	56.04	327.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.03	3.19	49.03	262.87	157,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	21.12	2,531,250.00 *gal*
Application event totals		26.03	3.19	49.03	283.99	

R4 - 10/17/2022: Oats, silage-soft dough

Field name: R4

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.96	61.65	172.03	6,473.54	242.00 *ton*
Application event totals		184.96	61.65	172.03	6,473.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	54.92	5,568,750.00 *gal*
Application event totals		0.00	0.00	0.00	54.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.93	4,556,250.00 *gal*
Application event totals		0.00	0.00	0.00	44.93	

R4 - 06/01/2023: Corn, silage

Field name: R4

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	158.40	43.56	182.16	15,166.80	242.00 *ton*
Application event totals		158.40	43.56	182.16	15,166.80	

R4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	47.93	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.95	3,543,750.00 *gal*
Application event totals		0.00	0.00	0.00	34.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	23.97	2,430,000.00 *gal*
Application event totals		80.00	0.00	0.00	23.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.96	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.97	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	24.96	2,531,250.00 *gal*
Application event totals		0.00	0.00	0.00	24.96	

R4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	22.97	2,328,750.00 *gal*
Application event totals		0.00	0.00	0.00	22.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	18.97	1,923,750.00 *gal*
Application event totals		0.00	0.00	0.00	18.97	

R5/R6 - 11/03/2022: Oats, silage-soft dough

Field name: R5/R6

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.96	61.65	172.03	6,473.54	99.00 *ton*
Application event totals		184.96	61.65	172.03	6,473.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	58.58	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	39.05	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

R5/R6 - 06/01/2023: Corn, silage

Field name: R5/R6

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	163.14	9.97	66.76	534.08	45,000.00 *gal*
Application event totals		163.14	9.97	66.76	534.08	

05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	73.23	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	73.23	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	41.50	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	41.50	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	24.41	1,012,500.00 *gal*
Application event totals		80.00	0.00	0.00	24.41	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	17.09	708,750.00 *gal*
Application event totals		0.00	0.00	0.00	17.09	

R5/R6 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.29	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	19.53	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	9.76	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.76	

R8 - 10/17/2022: Oats, silage-soft dough

Field name: R8

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.96	61.65	172.03	6,473.54	198.00 *ton*
Application event totals		184.96	61.65	172.03	6,473.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	61.02	5,062,500.00 *gal*
Application event totals		0.00	0.00	0.00	61.02	

R8 - 10/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	43.94	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.94	

R8 - 05/28/2023: Corn, silage

Field name: R8

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	158.40	43.56	182.16	15,166.80	198.00 *ton*
Application event totals		158.40	43.56	182.16	15,166.80	

05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	43.94	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.94	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	21.97	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	25.63	2,126,250.00 *gal*
Application event totals		80.00	0.00	0.00	25.63	

R8 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.39	2,936,250.00 *gal*
Application event totals		0.00	0.00	0.00	35.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	24.41	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	20.75	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	20.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	24.41	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.29	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

R9 - 10/16/2022: Oats, silage-soft dough

Field name: R9

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

R9 - 10/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.96	61.65	172.03	6,473.54	143.00 *ton*
Application event totals		184.96	61.65	172.03	6,473.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	70.97	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	70.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.04	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.04	

R9 - 05/28/2023: Corn, silage

Field name: R9

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	158.40	43.56	182.16	15,166.80	143.00 *ton*
Application event totals		158.40	43.56	182.16	15,166.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	43.94	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	43.94	

R9 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	23.66	1,417,500.00 *gal*
Application event totals		0.00	0.00	0.00	23.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	15.21	911,250.00 *gal*
Application event totals		80.00	0.00	0.00	15.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.49	2,126,250.00 *gal*
Application event totals		0.00	0.00	0.00	35.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	20.28	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	15.21	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	15.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	28.73	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	28.73	

R9 - 05/28/2023: Corn, silage

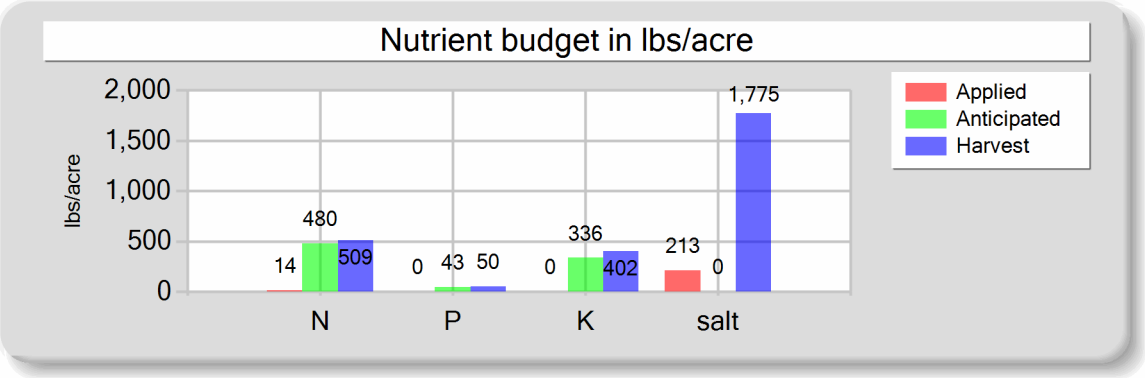
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	33.80	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.80	

B. NUTRIENT BUDGET

F1 - 01/01/2023: Alfalfa, hay

Field name: F1 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	212.74
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	212.74
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	509.04	50.40	401.52	1,774.94
Nutrient balance	-495.04	-50.40	-401.52	-1,562.19
Applied to removed ratio	0.03	0.00	0.00	0.12

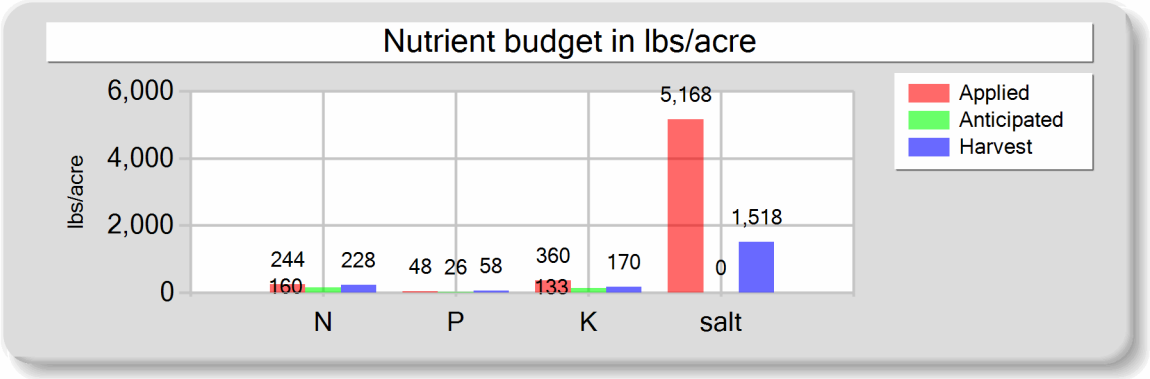
Fresh water applied	
18,630,000.00	*gallons*
686.08	*acre-inches*
36.11	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

F10 - 11/03/2022: Oats, silage-soft dough

Field name: F10 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.69	48.40	359.77	2,633.59
Fresh water	7.66	0.00	0.00	2,534.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.36	48.40	359.77	5,168.45
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	227.76	57.90	169.85	1,517.56
Nutrient balance	16.60	-9.51	189.91	3,650.89
Applied to removed ratio	1.07	0.84	2.12	3.41

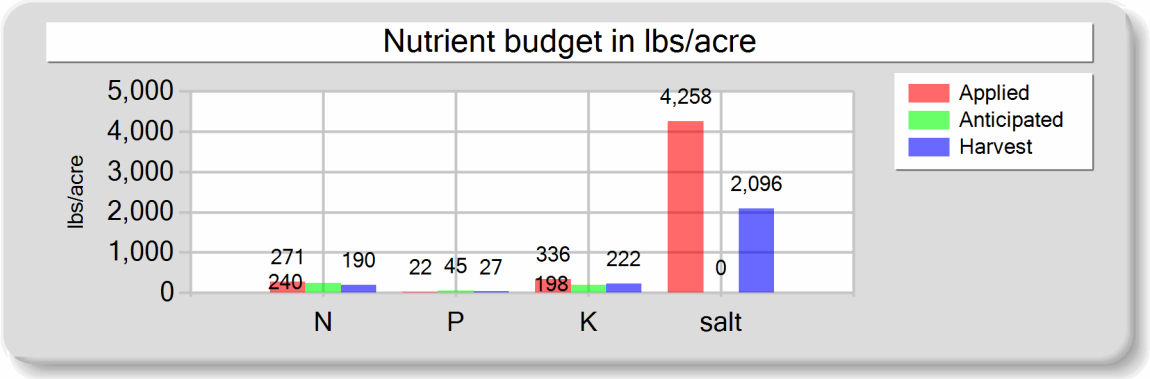
Fresh water applied	
6,376,500.00	*gallons*
234.82	*acre-inches*
18.06	*inches/acre*

Process wastewater applied	
1,215,000.00	*gallons*
44.74	*acre-inches*
3.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

F10 - 05/31/2023: Corn, silage

Field name: F10 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	178.52	21.91	336.24	1,802.52
Fresh water	5.78	0.00	0.00	2,455.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.30	21.91	336.24	4,257.77
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	189.63	27.09	222.14	2,096.12
Nutrient balance	81.67	-5.18	114.10	2,161.66
Applied to removed ratio	1.43	0.81	1.51	2.03

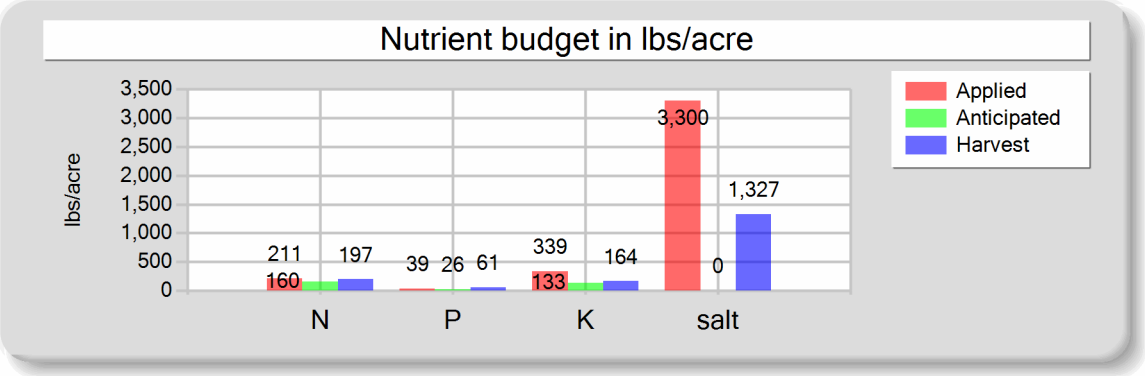
Fresh water applied	
12,701,250.00	*gallons*
467.74	*acre-inches*
35.98	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
1.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

F11 - 11/02/2022: Oats, silage-soft dough

Field name: F11 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.15	39.10	338.78	2,244.99
Fresh water	0.00	0.00	0.00	1,054.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.15	39.10	338.78	3,299.72
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	196.63	60.78	164.45	1,327.49
Nutrient balance	14.52	-21.67	174.33	1,972.23
Applied to removed ratio	1.07	0.64	2.06	2.49

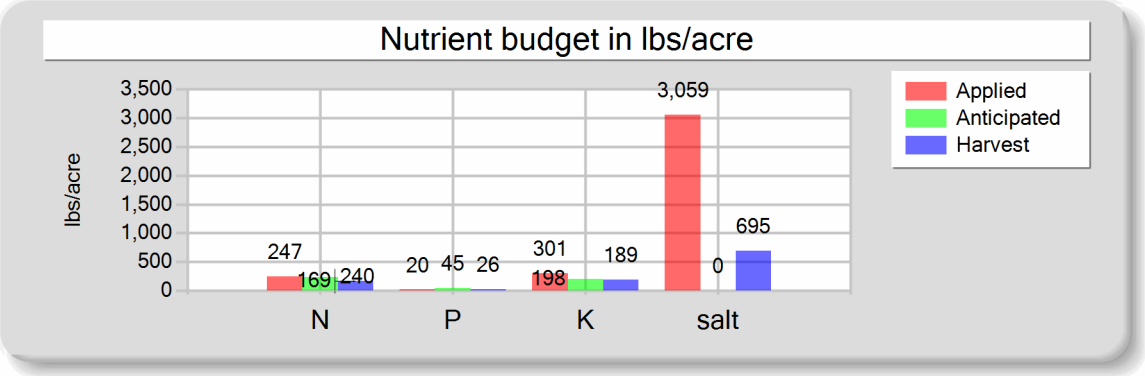
Fresh water applied	
46,530,000.00	*gallons*
1,713.54	*acre-inches*
59.09	*inches/acre*

Process wastewater applied	
2,205,000.00	*gallons*
81.20	*acre-inches*
2.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

F11 - 05/29/2023: Corn, silage

Field name: F11 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	160.05	19.64	301.46	1,616.05
Fresh water	0.00	0.00	0.00	1,442.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.05	19.64	301.46	3,058.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	168.71	25.56	189.16	694.68
Nutrient balance	78.34	-5.92	112.29	2,364.20
Applied to removed ratio	1.46	0.77	1.59	4.40

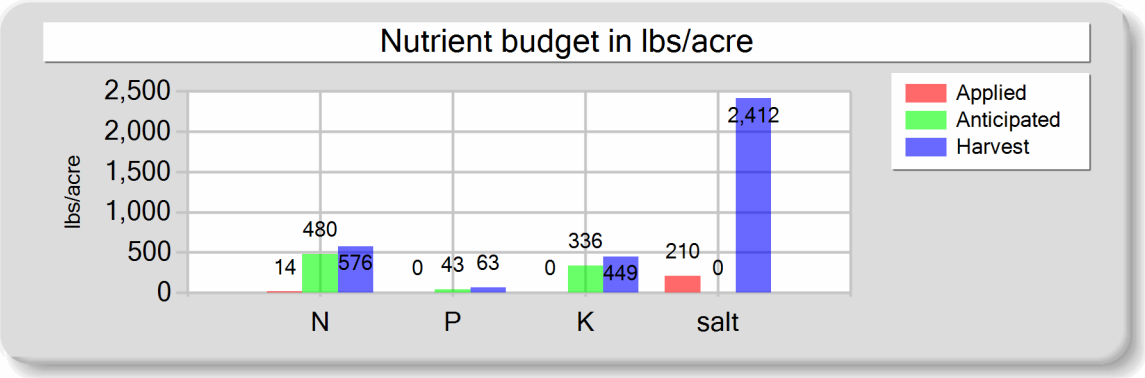
Fresh water applied	
28,597,500.00	*gallons*
1,053.15	*acre-inches*
36.32	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

F12 - 01/01/2023: Alfalfa, hay

Field name: F12 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	209.92
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	209.92
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	575.92	62.56	448.96	2,412.39
Nutrient balance	-561.92	-62.56	-448.96	-2,202.47
Applied to removed ratio	0.02	0.00	0.00	0.09

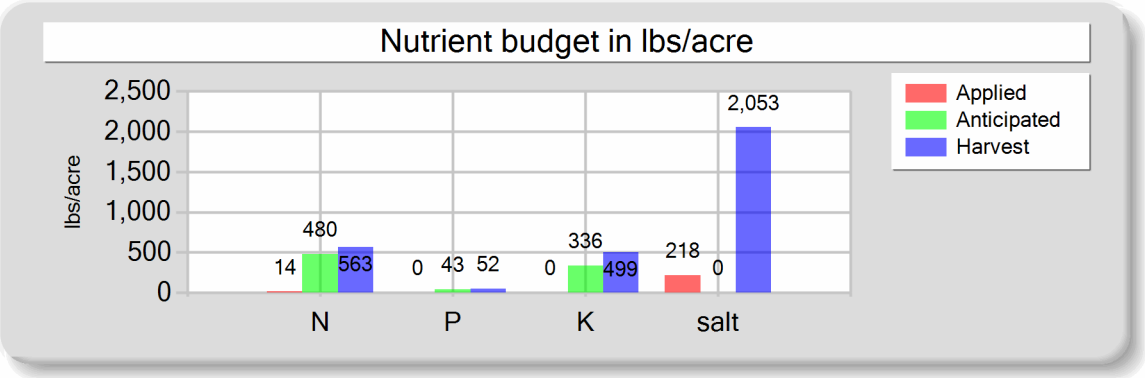
Fresh water applied	
26,122,500.00	*gallons*
962.00	*acre-inches*
35.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

F13 - 01/01/2023: Alfalfa, hay

Field name: F13 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	218.39
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	218.39
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	562.80	52.08	498.96	2,053.09
Nutrient balance	-548.80	-52.08	-498.96	-1,834.70
Applied to removed ratio	0.02	0.00	0.00	0.11

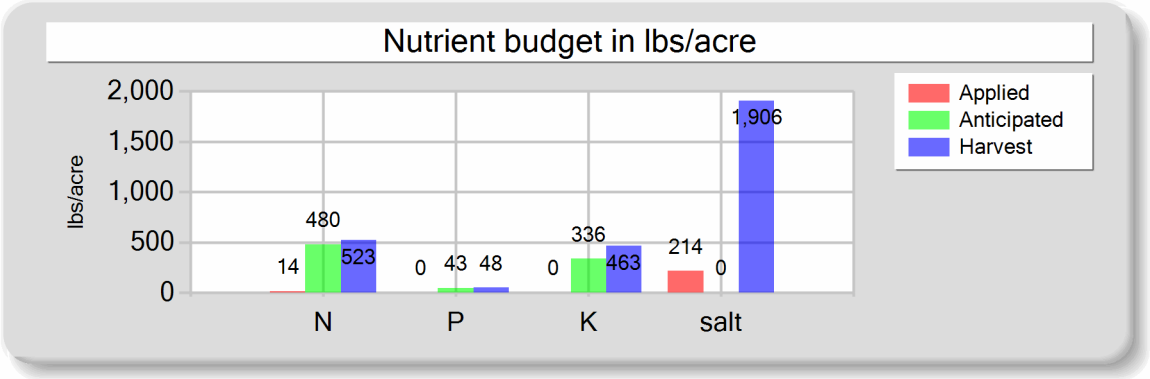
Fresh water applied	
17,111,250.00	*gallons*
630.15	*acre-inches*
37.07	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

F14 - 01/01/2023: Alfalfa, hay

Field name: F14 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	214.19
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	214.19
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	522.60	48.36	463.32	1,906.44
Nutrient balance	-508.60	-48.36	-463.32	-1,692.25
Applied to removed ratio	0.03	0.00	0.00	0.11

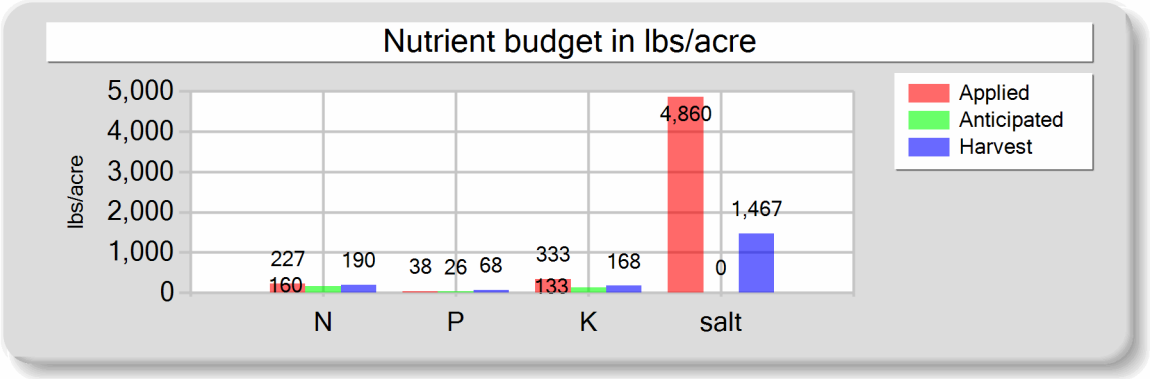
Fresh water applied
35,538,750.00 *gallons*
1,308.77 *acre-inches*
36.35 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

F2 - 10/16/2022: Oats, silage-soft dough

Field name: F2 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.60	37.71	333.30	2,382.88
Fresh water	12.38	0.00	0.00	2,477.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.97	37.71	333.30	4,860.12
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	189.84	68.05	168.35	1,467.47
Nutrient balance	37.14	-30.35	164.96	3,392.65
Applied to removed ratio	1.20	0.55	1.98	3.31

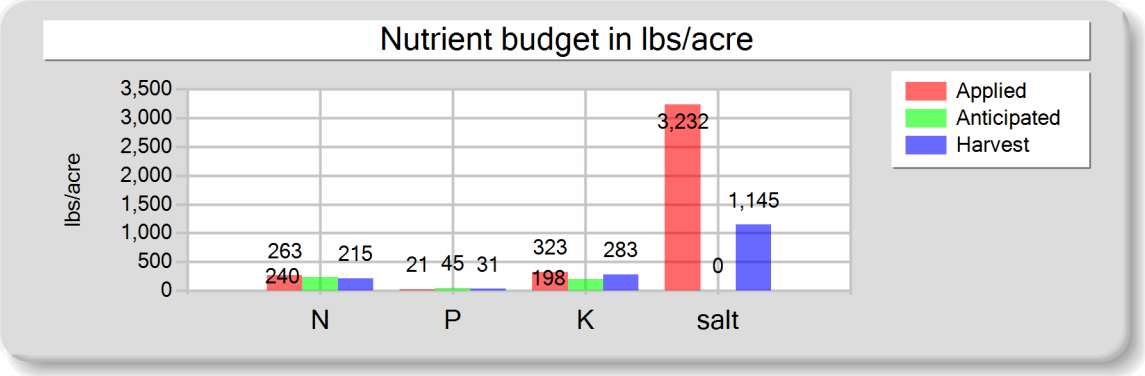
Fresh water applied	
9,705,000.00	*gallons*
357.40	*acre-inches*
16.25	*inches/acre*

Process wastewater applied	
1,552,500.00	*gallons*
57.17	*acre-inches*
2.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

F2 - 05/31/2023: Corn, silage

Field name: F2 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.42	21.04	322.87	1,730.83
Fresh water	4.57	0.00	0.00	1,501.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	262.99	21.04	322.87	3,231.95
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	215.15	30.74	282.77	1,144.97
Nutrient balance	47.83	-9.70	40.10	2,086.98
Applied to removed ratio	1.22	0.68	1.14	2.82

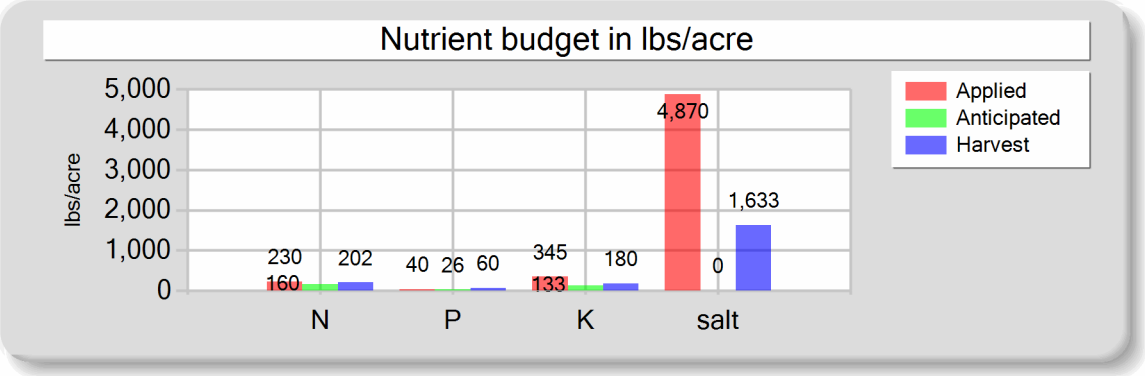
Fresh water applied	
22,692,750.00	*gallons*
835.70	*acre-inches*
37.99	*inches/acre*

Process wastewater applied	
877,500.00	*gallons*
32.32	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

F3 - 11/04/2022: Oats, silage-soft dough

Field name: F3 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	211.08	40.21	344.77	2,362.38
Fresh water	12.24	0.00	0.00	2,508.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.32	40.21	344.77	4,870.46
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	201.92	60.15	180.44	1,632.81
Nutrient balance	28.41	-19.93	164.34	3,237.65
Applied to removed ratio	1.14	0.67	1.91	2.98

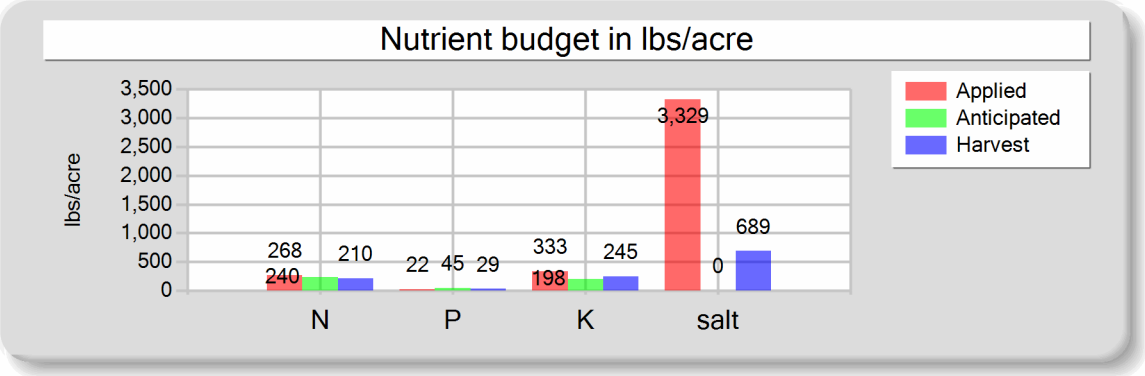
Fresh water applied	
10,467,000.00	*gallons*
385.46	*acre-inches*
16.76	*inches/acre*

Process wastewater applied	
1,777,500.00	*gallons*
65.46	*acre-inches*
2.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

F3 - 06/01/2023: Corn, silage

Field name: F3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	176.58	21.67	332.58	1,782.93
Fresh water	4.78	0.00	0.00	1,545.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.35	21.67	332.58	3,328.70
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	209.95	29.16	244.94	688.64
Nutrient balance	58.40	-7.49	87.64	2,640.05
Applied to removed ratio	1.28	0.74	1.36	4.83

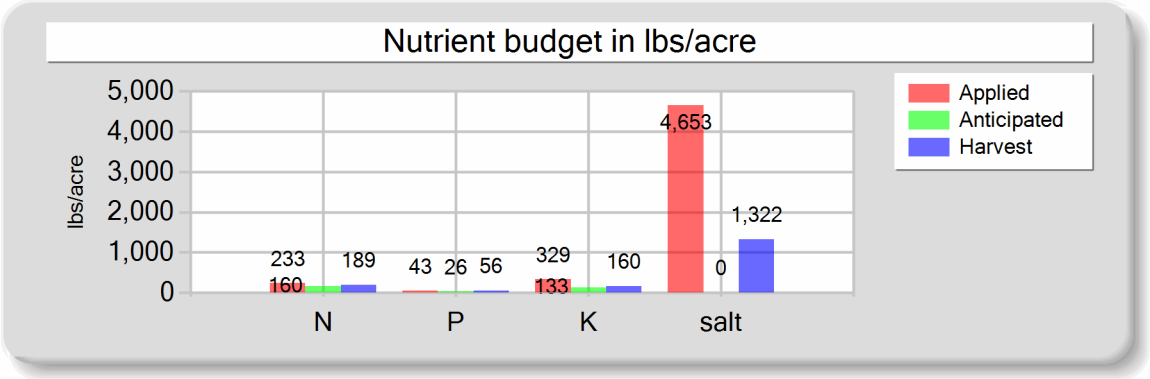
Fresh water applied	
22,296,000.00	*gallons*
821.09	*acre-inches*
35.70	*inches/acre*

Process wastewater applied	
945,000.00	*gallons*
34.80	*acre-inches*
1.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

F4 - 11/02/2022: Oats, silage-soft dough

Field name: F4 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.61	43.04	329.30	2,506.42
Fresh water	12.40	0.00	0.00	2,146.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.02	43.04	329.30	4,653.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	189.21	55.65	159.53	1,322.49
Nutrient balance	43.80	-12.61	169.77	3,330.52
Applied to removed ratio	1.23	0.77	2.06	3.52

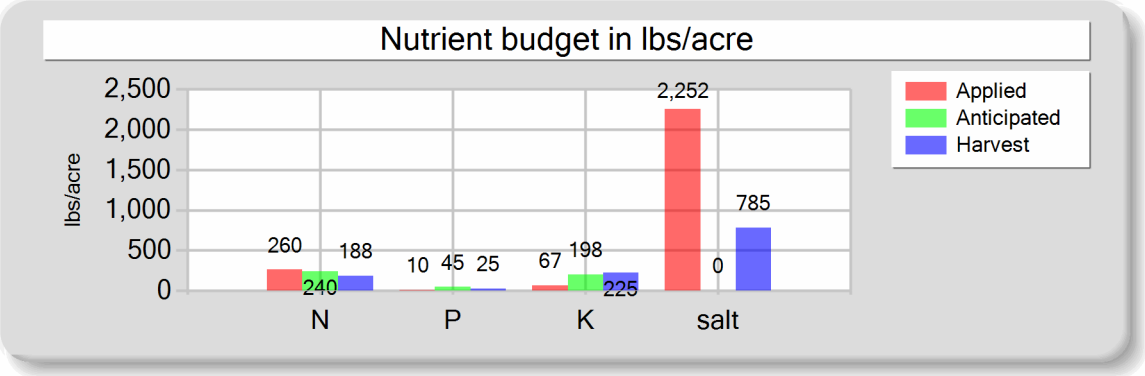
Fresh water applied	
14,703,000.00	*gallons*
541.46	*acre-inches*
16.92	*inches/acre*

Process wastewater applied	
2,599,405.00	*gallons*
95.73	*acre-inches*
2.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

F4 - 06/01/2023: Corn, silage

Field name: F4 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.14	9.97	66.76	534.08
Fresh water	9.97	0.00	0.00	1,717.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	260.12	9.97	66.76	2,251.75
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	187.62	25.02	225.14	784.50
Nutrient balance	72.50	-15.04	-158.38	1,467.25
Applied to removed ratio	1.39	0.40	0.30	2.87

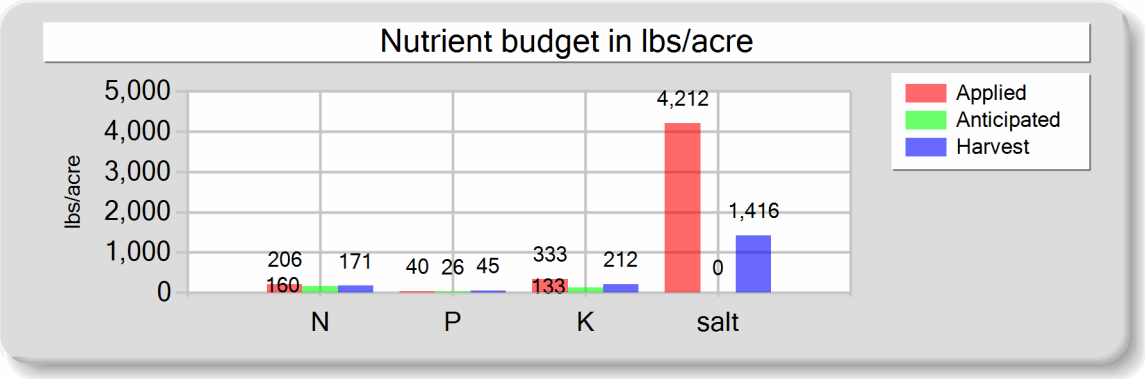
Fresh water applied	
31,128,750.00	*gallons*
1,146.37	*acre-inches*
35.82	*inches/acre*

Process wastewater applied	
160,000.00	*gallons*
5.89	*acre-inches*
0.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

F5 - 11/02/2022: Oats, silage-soft dough

Field name: F5 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	198.97	40.17	332.50	2,150.92
Fresh water	0.00	0.00	0.00	2,061.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.97	40.17	332.50	4,212.33
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	170.88	44.75	211.57	1,416.03
Nutrient balance	35.09	-4.58	120.93	2,796.31
Applied to removed ratio	1.21	0.90	1.57	2.97

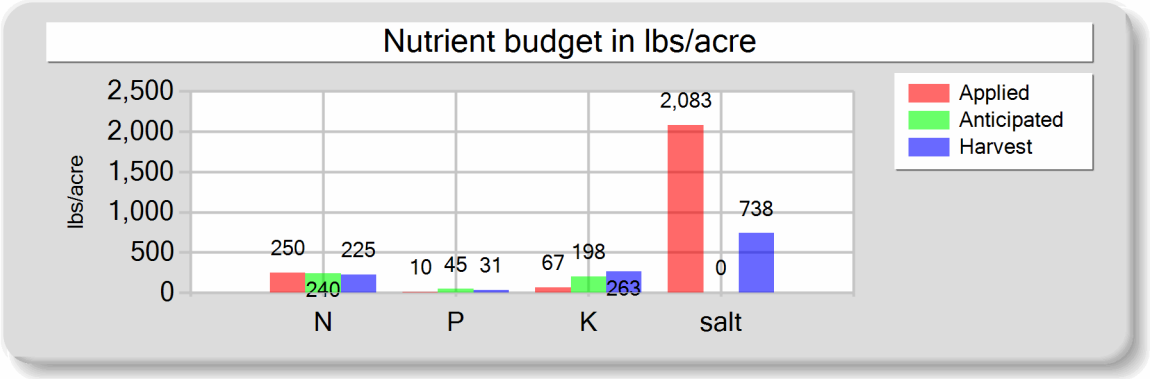
Fresh water applied	
9,262,500.00	*gallons*
341.11	*acre-inches*
16.24	*inches/acre*

Process wastewater applied	
1,671,551.00	*gallons*
61.56	*acre-inches*
2.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

F5 - 05/30/2023: Corn, silage

Field name: F5 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.14	9.97	66.76	534.08
Fresh water	0.00	0.00	0.00	1,548.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.14	9.97	66.76	2,082.90
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.14	31.27	262.67	738.47
Nutrient balance	25.00	-21.30	-195.91	1,344.42
Applied to removed ratio	1.11	0.32	0.25	2.82

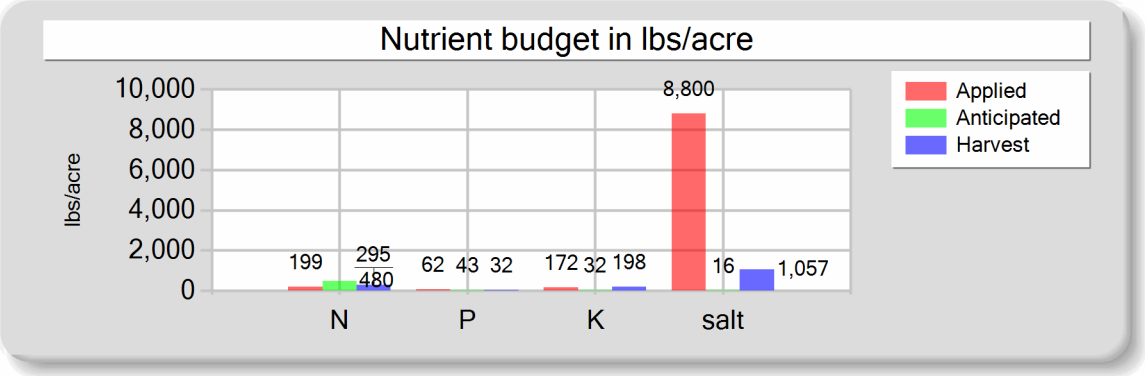
Fresh water applied	
21,243,750.00	*gallons*
782.34	*acre-inches*
37.25	*inches/acre*

Process wastewater applied	
105,000.00	*gallons*
3.87	*acre-inches*
0.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

F6 - 01/01/2023: Alfalfa, hay

Field name: F6 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.96	61.65	172.03	6,473.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,325.99
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	198.96	61.65	172.03	8,799.53
Anticipated crop nutrient removal	480.00	43.20	32.00	16.00
Actual crop nutrient removal	295.20	32.40	198.00	1,056.51
Nutrient balance	-96.24	29.25	-25.97	7,743.02
Applied to removed ratio	0.67	1.90	0.87	8.33

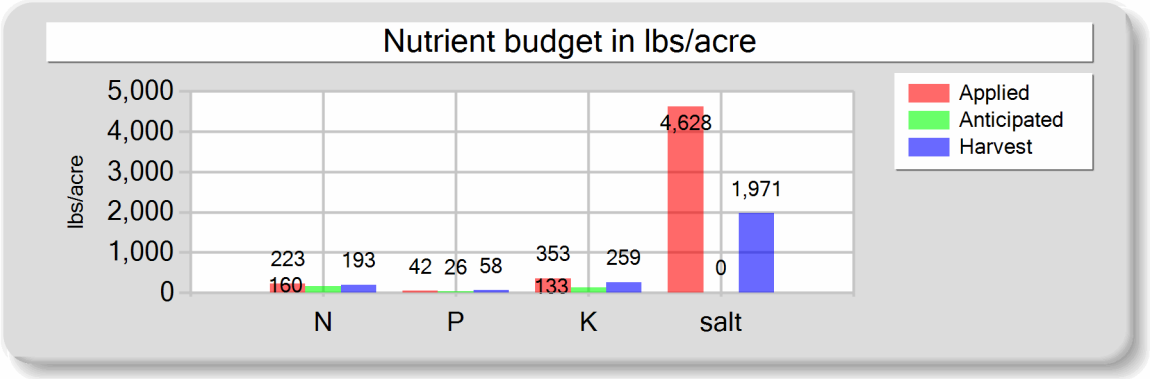
Fresh water applied	
6,150,000.00	*gallons*
226.48	*acre-inches*
37.75	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

F7 - 10/16/2022: Oats, silage-soft dough

Field name: F7 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.95	41.81	353.05	2,284.65
Fresh water	4.78	0.00	0.00	2,343.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.73	41.81	353.05	4,627.86
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	193.45	57.62	259.31	1,970.74
Nutrient balance	29.28	-15.82	93.74	2,657.11
Applied to removed ratio	1.15	0.73	1.36	2.35

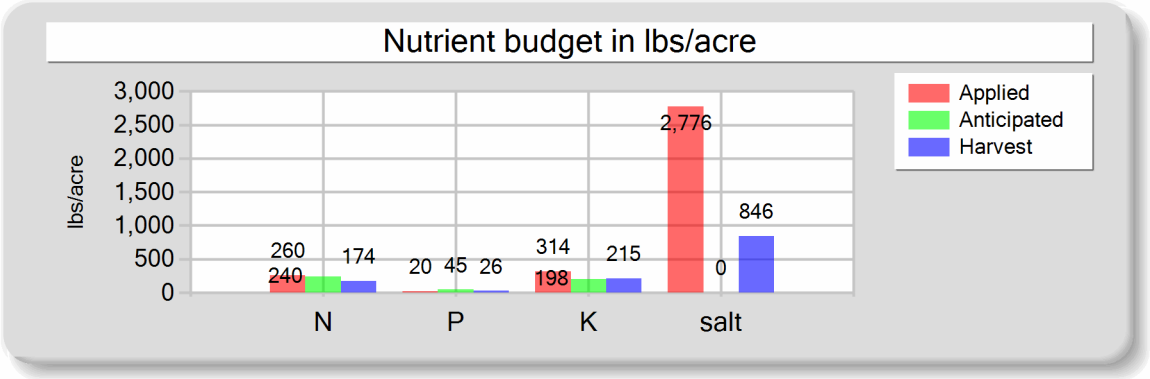
Fresh water applied	
8,899,500.00	*gallons*
327.74	*acre-inches*
18.21	*inches/acre*

Process wastewater applied	
1,485,000.00	*gallons*
54.69	*acre-inches*
3.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

F7 - 05/30/2023: Corn, silage

Field name: F7 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	166.53	20.44	313.67	1,681.52
Fresh water	6.89	0.00	0.00	1,094.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	260.43	20.44	313.67	2,776.24
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	174.04	25.59	214.99	845.83
Nutrient balance	86.39	-5.16	98.68	1,930.41
Applied to removed ratio	1.50	0.80	1.46	3.28

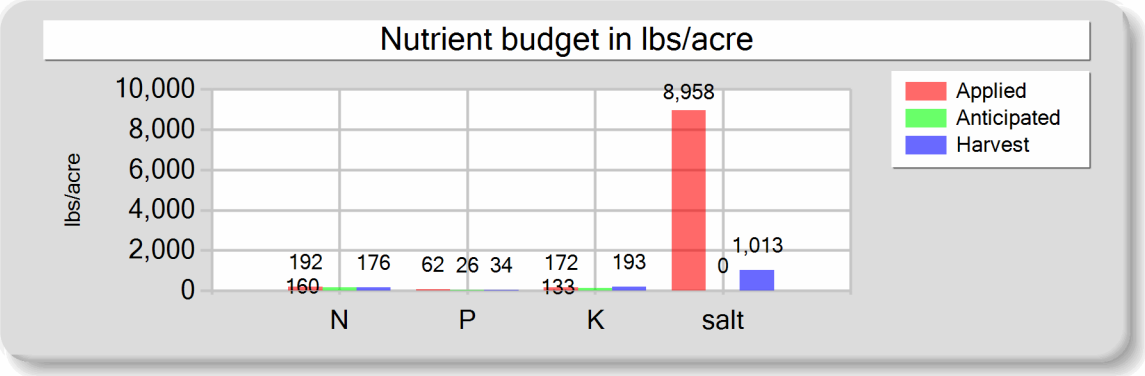
Fresh water applied	
17,968,500.00	*gallons*
661.72	*acre-inches*
36.76	*inches/acre*

Process wastewater applied	
697,500.00	*gallons*
25.69	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

F8 - 10/17/2022: Oats, silage-soft dough

Field name: F8 Crop: Oats, silage-soft dough Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.96	61.65	172.03	6,473.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,484.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.96	61.65	172.03	8,957.62
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	176.45	33.61	193.25	1,013.19
Nutrient balance	15.51	28.04	-21.22	7,944.43
Applied to removed ratio	1.09	1.83	0.89	8.84

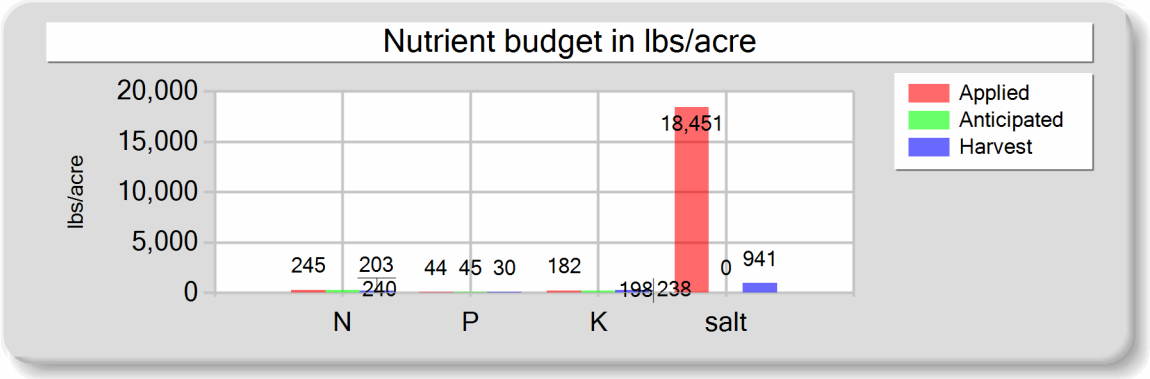
Fresh water applied	
28,440,000.00	*gallons*
1,047.35	*acre-inches*
18.37	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

F8 - 05/29/2023: Corn, silage

Field name: F8 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	158.40	43.56	182.16	15,166.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	3,284.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.40	43.56	182.16	18,451.24
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	202.63	29.80	238.39	941.28
Nutrient balance	42.77	13.76	-56.23	17,509.96
Applied to removed ratio	1.21	1.46	0.76	19.60

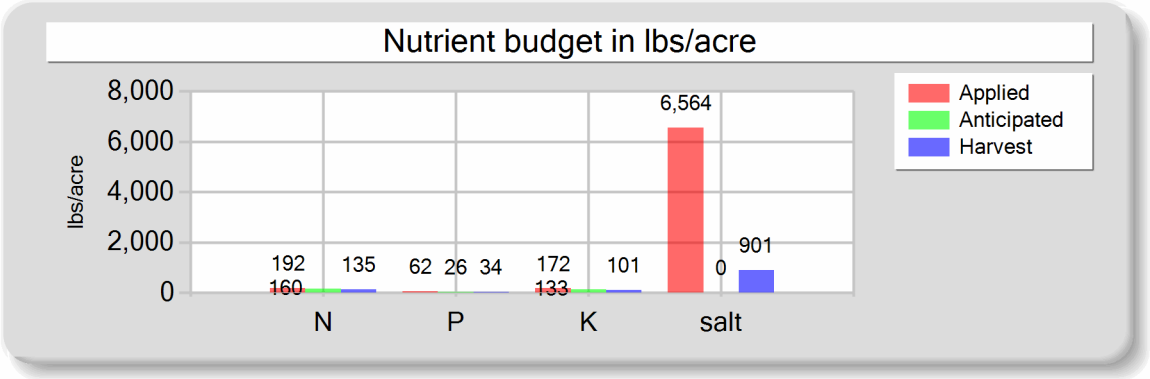
Fresh water applied	
54,742,500.00	*gallons*
2,015.98	*acre-inches*
35.37	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

F9 - 11/03/2022: Oats, silage-soft dough

Field name: F9 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.96	61.65	172.03	6,473.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	90.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.96	61.65	172.03	6,563.86
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	135.29	34.40	100.90	901.47
Nutrient balance	56.66	27.26	71.13	5,662.39
Applied to removed ratio	1.42	1.79	1.71	7.28

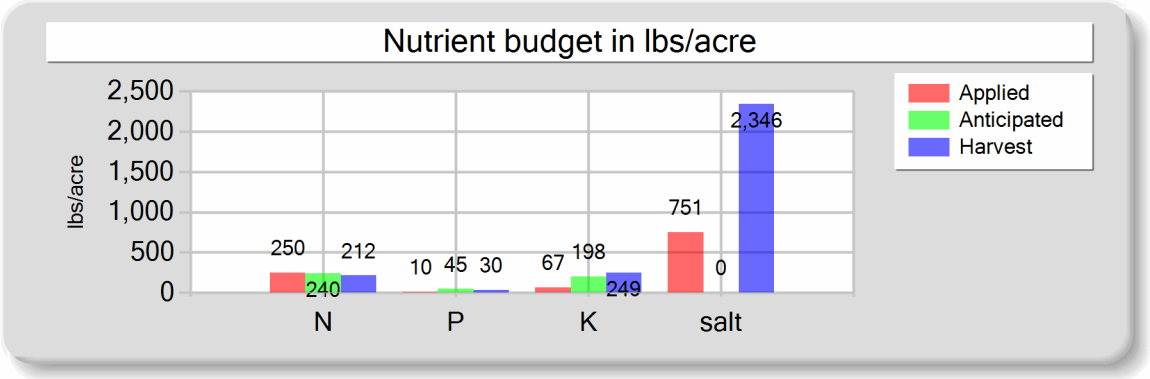
Fresh water applied	
7,492,500.00	*gallons*
275.92	*acre-inches*
15.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

F9 - 05/31/2023: Corn, silage

Field name: F9 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.14	9.97	66.76	534.08
Fresh water	0.00	0.00	0.00	217.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.14	9.97	66.76	751.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.23	30.32	248.61	2,345.95
Nutrient balance	37.91	-20.35	-181.85	-1,594.63
Applied to removed ratio	1.18	0.33	0.27	0.32

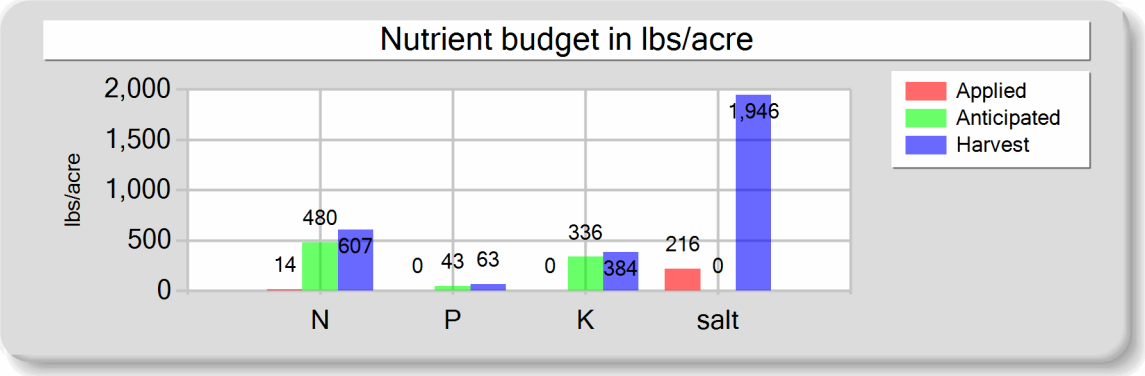
Fresh water applied	
18,022,500.00	*gallons*
663.71	*acre-inches*
36.87	*inches/acre*

Process wastewater applied	
90,000.00	*gallons*
3.31	*acre-inches*
0.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

R1 - 01/01/2023: Alfalfa, hay

Field name: R1 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	215.69
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	215.69
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	607.20	63.36	383.68	1,946.45
Nutrient balance	-593.20	-63.36	-383.68	-1,730.77
Applied to removed ratio	0.02	0.00	0.00	0.11

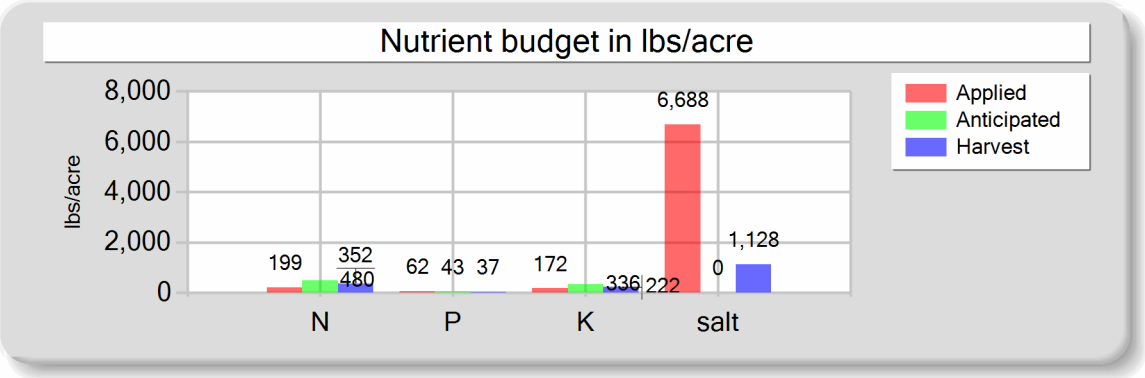
Fresh water applied	
21,870,000.00	*gallons*
805.40	*acre-inches*
36.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

R2 - 01/01/2023: Alfalfa, hay

Field name: R2 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.96	61.65	172.03	6,473.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	214.19
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	198.96	61.65	172.03	6,687.73
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	351.90	36.72	222.36	1,128.06
Nutrient balance	-152.94	24.93	-50.33	5,559.68
Applied to removed ratio	0.57	1.68	0.77	5.93

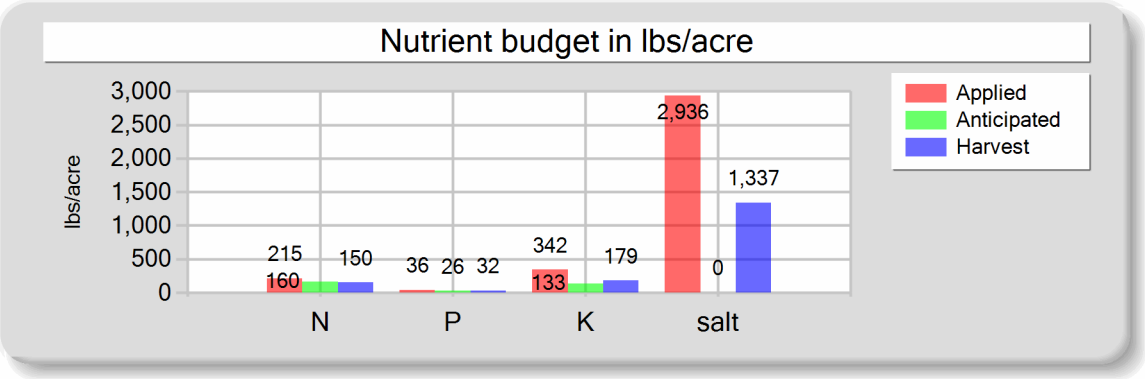
Fresh water applied	
11,846,250.00	*gallons*
436.26	*acre-inches*
36.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

R3 - 10/16/2022: Oats, silage-soft dough

Field name: R3 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.42	36.11	341.67	2,236.40
Fresh water	4.73	0.00	0.00	699.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.15	36.11	341.67	2,936.11
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	150.20	31.96	178.96	1,337.05
Nutrient balance	64.95	4.16	162.71	1,599.07
Applied to removed ratio	1.43	1.13	1.91	2.20

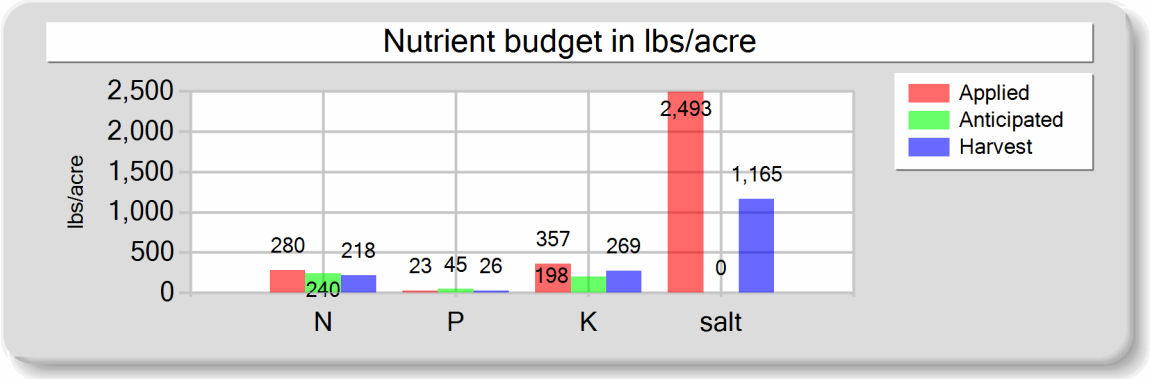
Fresh water applied	
11,655,000.00	*gallons*
429.21	*acre-inches*
16.51	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
2.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

R3 - 06/01/2023: Corn, silage

Field name: R3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	189.68	23.28	357.25	1,915.18
Fresh water	2.84	0.00	0.00	577.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.51	23.28	357.25	2,492.50
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.90	25.64	269.17	1,164.87
Nutrient balance	61.61	-2.36	88.08	1,327.63
Applied to removed ratio	1.28	0.91	1.33	2.14

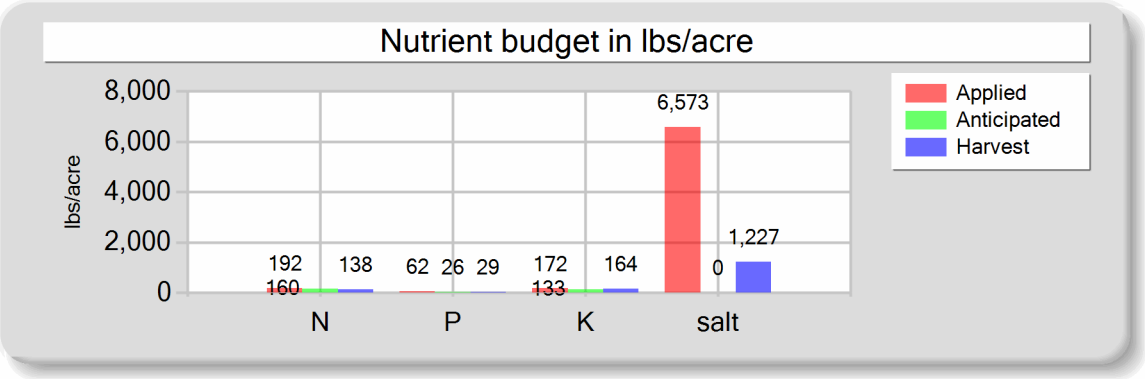
Fresh water applied	
25,866,000.00	*gallons*
952.56	*acre-inches*
36.64	*inches/acre*

Process wastewater applied	
1,147,500.00	*gallons*
42.26	*acre-inches*
1.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

R4 - 10/17/2022: Oats, silage-soft dough

Field name: R4 Crop: Oats, silage-soft dough Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.96	61.65	172.03	6,473.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	99.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.96	61.65	172.03	6,573.40
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	137.87	29.33	164.27	1,227.30
Nutrient balance	54.09	32.32	7.76	5,346.10
Applied to removed ratio	1.39	2.10	1.05	5.36

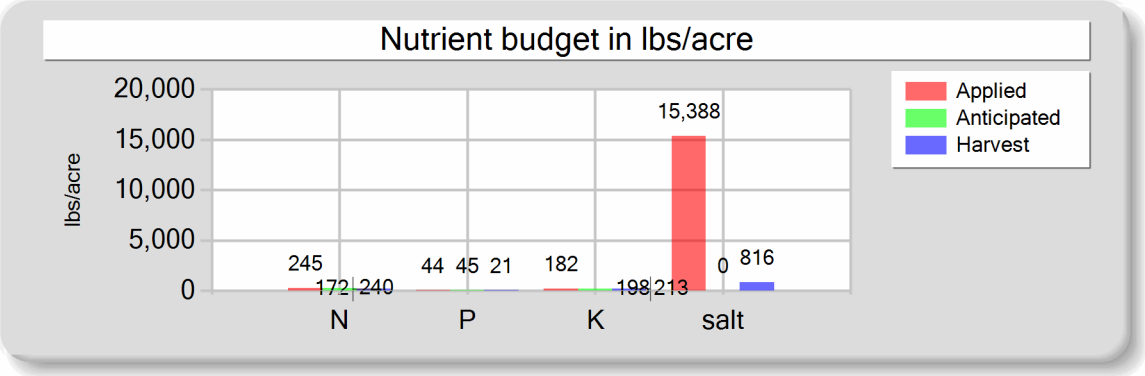
Fresh water applied	
10,125,000.00	*gallons*
372.87	*acre-inches*
16.95	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

R4 - 06/01/2023: Corn, silage

Field name: R4 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	158.40	43.56	182.16	15,166.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	221.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.40	43.56	182.16	15,388.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	171.68	20.81	213.29	816.44
Nutrient balance	73.73	22.75	-31.13	14,572.03
Applied to removed ratio	1.43	2.09	0.85	18.85

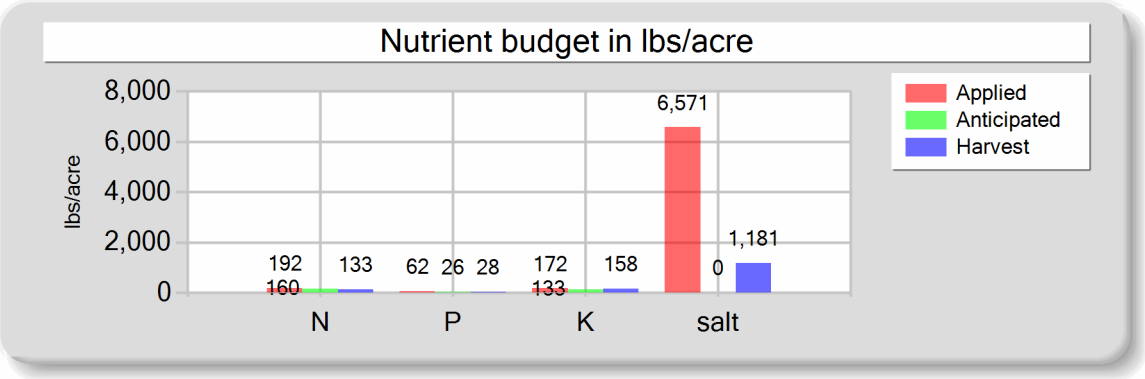
Fresh water applied	
22,477,500.00	*gallons*
827.77	*acre-inches*
37.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

R5/R6 - 11/03/2022: Oats, silage-soft dough

Field name: R5/R6 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.96	61.65	172.03	6,473.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	97.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.96	61.65	172.03	6,571.18
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.64	28.22	158.04	1,180.76
Nutrient balance	59.31	33.43	13.99	5,390.42
Applied to removed ratio	1.45	2.18	1.09	5.57

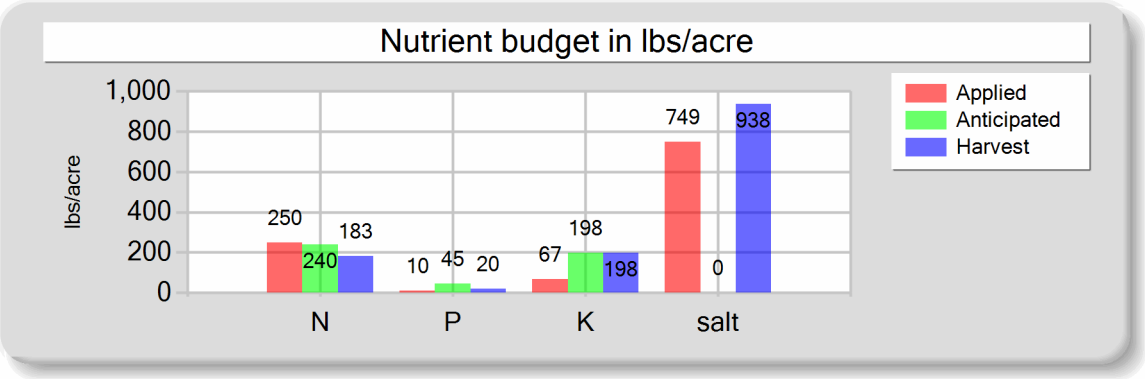
Fresh water applied	
4,050,000.00	*gallons*
149.15	*acre-inches*
16.57	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

R5/R6 - 06/01/2023: Corn, silage

Field name: R5/R6 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.14	9.97	66.76	534.08
Fresh water	0.00	0.00	0.00	214.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.14	9.97	66.76	748.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	182.76	20.31	197.99	938.17
Nutrient balance	67.38	-10.33	-131.23	-189.29
Applied to removed ratio	1.37	0.49	0.34	0.80

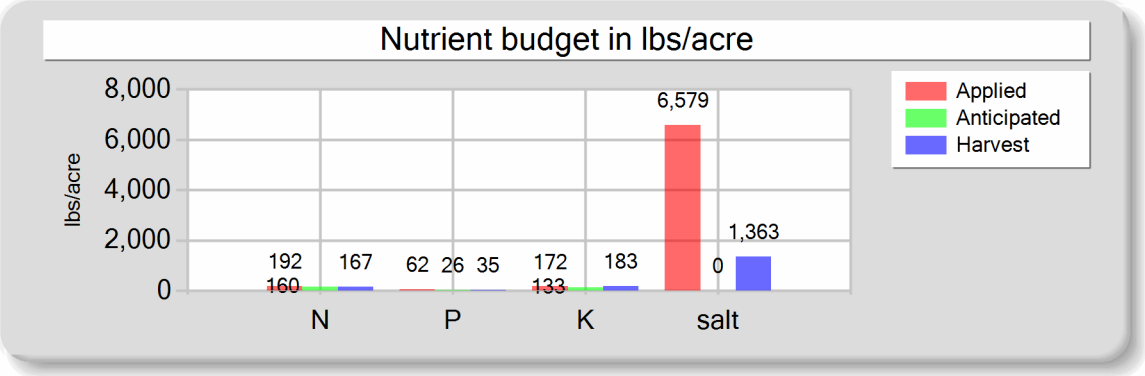
Fresh water applied
8,910,000.00 *gallons*
328.13 *acre-inches*
36.46 *inches/acre*

Process wastewater applied
45,000.00 *gallons*
1.66 *acre-inches*
0.18 *inches/acre*

Total harvests for the crop
1 *harvests*

R8 - 10/17/2022: Oats, silage-soft dough

Field name: R8 Crop: Oats, silage-soft dough Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.96	61.65	172.03	6,473.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	104.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.96	61.65	172.03	6,578.50
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	166.98	35.32	183.03	1,362.64
Nutrient balance	24.98	26.33	-11.00	5,215.87
Applied to removed ratio	1.15	1.75	0.94	4.83

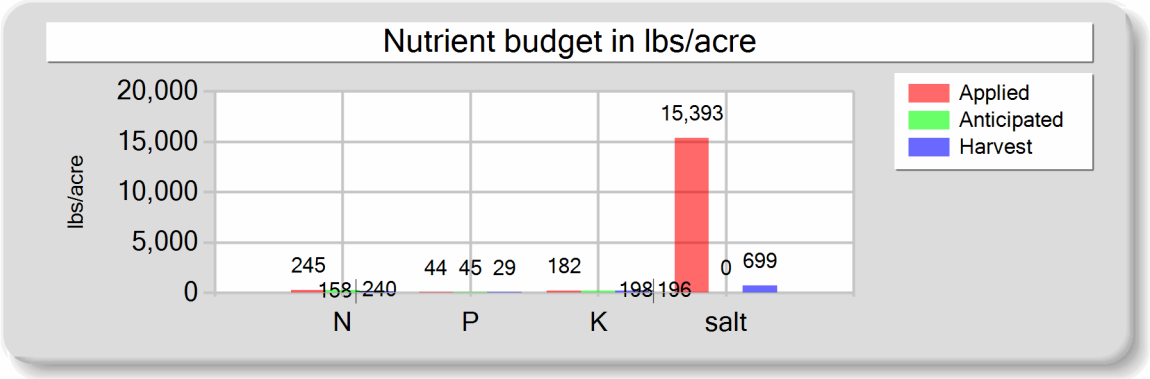
Fresh water applied	
8,707,500.00	*gallons*
320.67	*acre-inches*
17.81	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

R8 - 05/28/2023: Corn, silage

Field name: R8 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	158.40	43.56	182.16	15,166.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	225.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.40	43.56	182.16	15,392.58
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	157.76	28.68	196.00	699.30
Nutrient balance	87.64	14.88	-13.84	14,693.28
Applied to removed ratio	1.56	1.52	0.93	22.01

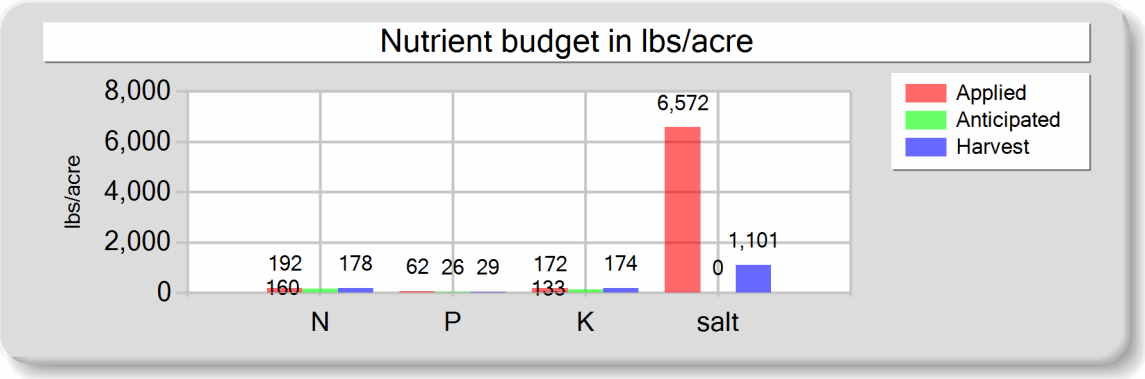
Fresh water applied	
18,731,250.00	*gallons*
689.81	*acre-inches*
38.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

R9 - 10/16/2022: Oats, silage-soft dough

Field name: R9 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.96	61.65	172.03	6,473.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	98.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.96	61.65	172.03	6,571.56
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	177.69	29.08	174.46	1,101.24
Nutrient balance	14.27	32.58	-2.43	5,470.32
Applied to removed ratio	1.08	2.12	0.99	5.97

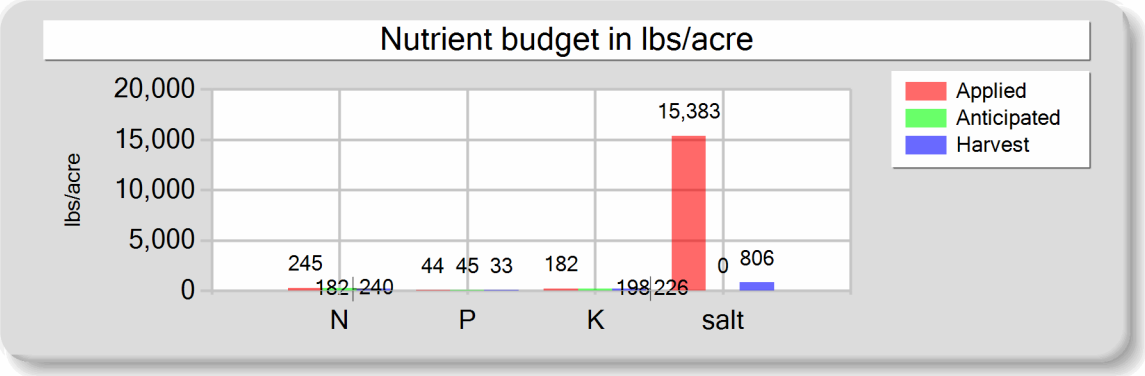
Fresh water applied	
5,872,500.00	*gallons*
216.26	*acre-inches*
16.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

R9 - 05/28/2023: Corn, silage

Field name: R9 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	158.40	43.56	182.16	15,166.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	216.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.40	43.56	182.16	15,383.10
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	181.80	33.06	225.88	805.89
Nutrient balance	63.60	10.50	-43.72	14,577.21
Applied to removed ratio	1.35	1.32	0.81	19.09

Fresh water applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
36.71	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Separated Solids

Sample and source description: 2022 Fall Separated Solids

Sample date: 12/05/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	6,200.00	17,300.00	17,500.00	9,700.00	3,800.00	4,000.00	1,220.00		65.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,000.00	2,200.00	9,200.00							76.60
DL	100.00	100.00	100.00							0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 85.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	1,700.00	6,200.00							17.00
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	3,900.00	10,400.00	8,800.00	3,900.00	4,700.00	2,200.00	773.00		21.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	476.00	260.00	0.00	0.00	74.70	677.00								7,400.00	6,200
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	179.00	118.00	0.00	0.00	54.00	324.00								2,900.00	1,580
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 05/01/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3,910.00	1,270.00	0.00	0.00	239.00	1,600.00								17,500.00	12,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	515.00	263.00	0.00	0.00	63.20	970.00								8,210.00	5,200
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,640.00	280.00	0.00	0.00	71.00	473.00	104.00	99.50	543.00	3,180.00	0.00	165.00	415.00	5,280.00	2,260
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	636.00	333.00	0.00	0.00	53.70	770.00								8,220.00	8,440
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

Sample description: Canal

Sample date: 08/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.90	26
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

Alberto

Sample description: Alberto

Sample date: 10/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.72	0.00	0.92								1,030.00	500
DL	0.50	0.50	0.50								1.00	10

New Well #1

Sample description: New Well #1

Sample date: 10/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.25	0.00	0.64								2,020.00	970
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

New Well #2

Sample description: New Well #2

Sample date: 10/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								2,010.00	975
DL	0.50	0.50	0.50								1.00	10

R & J

Sample description: R & J

Sample date: 10/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								2,000.00	975
DL	0.50	0.50	0.50								1.00	10

Tower

Sample description: Tower

Sample date: 10/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.59	0.00	0.79								2,010.00	1,030
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

F1 - 01/01/2023: Alfalfa, hay

F1 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,300.00	3,000.00	23,900.00		11.70
DL	100.00	100.00	100.00		0.01

F10 - 11/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	1,500.00	4,400.00		9.36
DL	100.00	100.00	100.00		0.01

F10 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	500.00	4,100.00		12.40
DL	100.00	100.00	100.00		0.01

F11 - 11/02/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,700.00	4,600.00		8.82
DL	100.00	100.00	100.00		0.01

F11 - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	500.00	3,700.00		4.30
DL	100.00	100.00	100.00		0.01

F12 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,300.00	3,400.00	24,400.00		14.60
DL	100.00	100.00	100.00		0.01

F13 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,500.00	3,100.00	29,700.00		13.40
DL	100.00	100.00	100.00		0.01

F14 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,500.00	3,100.00	29,700.00		13.40
DL	100.00	100.00	100.00		0.01

F2 - 10/16/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,300.00	1,900.00	4,700.00		9.64
DL	100.00	100.00	100.00		0.01

F2 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	500.00	4,600.00		6.70
DL	100.00	100.00	100.00		0.01

F3 - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	1,400.00	4,200.00		10.30
DL	100.00	100.00	100.00		0.01

F3 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	500.00	4,200.00		4.10
DL	100.00	100.00	100.00		0.01

F4 - 11/02/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	1,500.00	4,300.00		9.14
DL	100.00	100.00	100.00		0.01

F4 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,000.00	400.00	3,600.00		4.90
DL	100.00	100.00	100.00		0.01

F5 - 11/02/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	1,100.00	5,200.00		11.30
DL	100.00	100.00	100.00		0.01

F6 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,800.00	3,600.00	22,000.00		13.00
DL	100.00	100.00	100.00		0.01

F7 - 10/16/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	1,400.00	6,300.00		12.00
DL	100.00	100.00	100.00		0.01

F7 - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	500.00	4,200.00		5.40
DL	100.00	100.00	100.00		0.01

F8 - 10/17/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,300.00	1,200.00	6,900.00		11.20
DL	100.00	100.00	100.00		0.01

F8 - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	500.00	4,000.00		5.30
DL	100.00	100.00	100.00		0.01

F9 - 11/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	1,500.00	4,400.00		9.36
DL	100.00	100.00	100.00		0.01

R1 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,500.00	3,600.00	21,800.00		12.10
DL	100.00	100.00	100.00		0.01

R2 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 08/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,500.00	3,600.00	21,800.00		12.10
DL	100.00	100.00	100.00		0.01

R3 - 10/16/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	1,000.00	5,600.00		11.40
DL	100.00	100.00	100.00		0.01

R3 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	400.00	4,200.00		6.40
DL	100.00	100.00	100.00		0.01

R4 - 10/17/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	1,000.00	5,600.00		11.40
DL	100.00	100.00	100.00		0.01

R4 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	400.00	4,100.00		5.90
DL	100.00	100.00	100.00		0.01

R4 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	500.00	3,800.00		4.80
DL	100.00	100.00	100.00		0.01

R5/R6 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	400.00	3,900.00		5.60
DL	100.00	100.00	100.00		0.01

R8 - 10/17/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,100.00	5,700.00		12.30
DL	100.00	100.00	100.00		0.01

R8 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	600.00	4,100.00		4.60
DL	100.00	100.00	100.00		0.01

R9 - 10/16/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	900.00	5,400.00		11.40
DL	100.00	100.00	100.00		0.01

R9 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,800.00	500.00	3,900.00		5.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

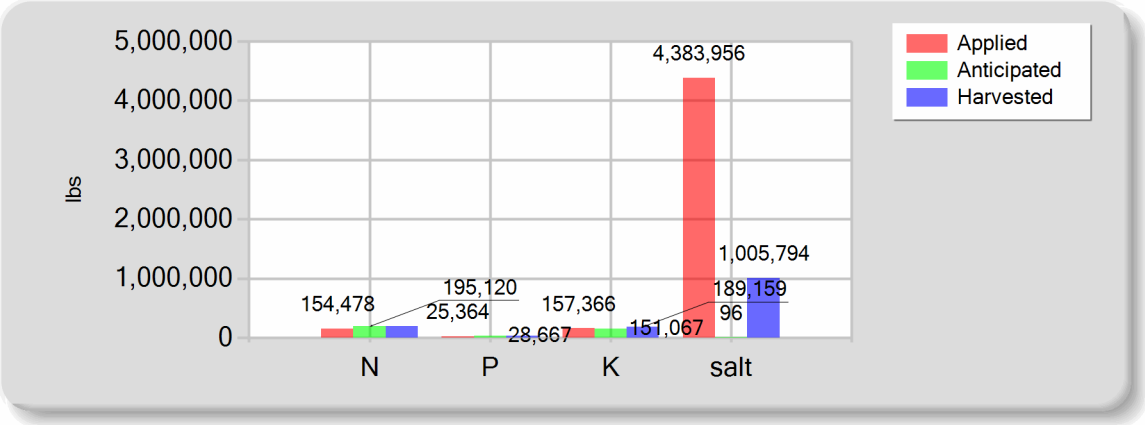
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

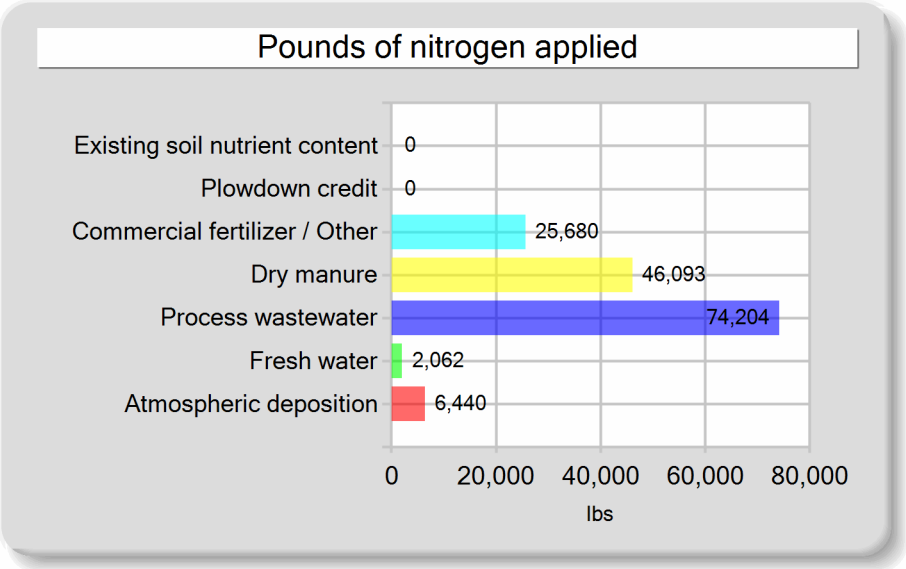
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

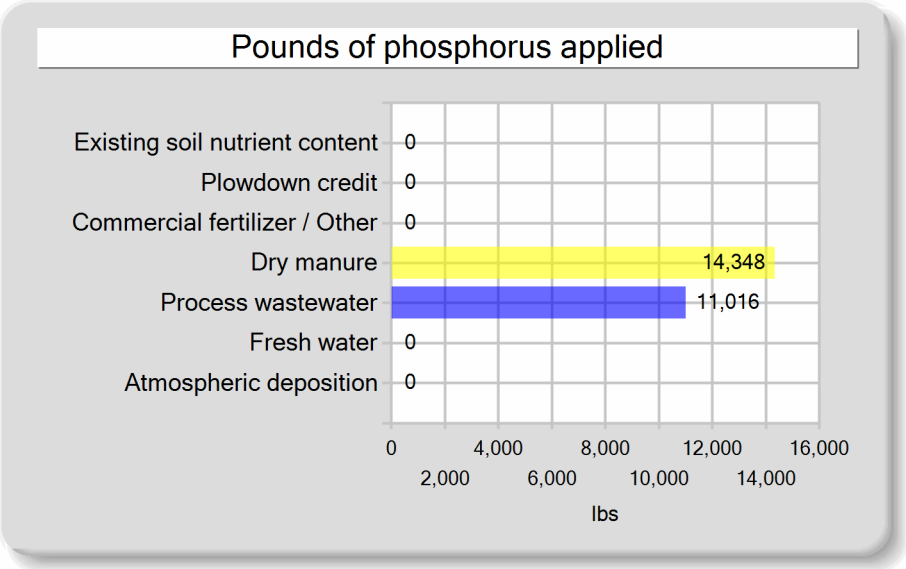
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25,680.00	0.00	0.00	0.00
Dry manure	46,092.55	14,347.78	46,702.44	2,671,747.32
Process wastewater	74,203.78	11,016.42	110,663.87	702,916.55
Fresh water	2,061.78	0.00	0.00	1,009,291.69
Atmospheric deposition	6,440.00	0.00	0.00	0.00
Total nutrients applied	154,478.11	25,364.20	157,366.31	4,383,955.56
Anticipated crop nutrient removal	195,120.00	28,667.40	151,066.80	96.00
Actual crop nutrient removal	191,949.72	30,590.19	189,158.88	1,005,794.11
Nutrient balance	-37,471.60	-5,225.99	-31,792.57	3,378,161.45
Applied to removed ratio	0.80	0.83	0.83	4.36

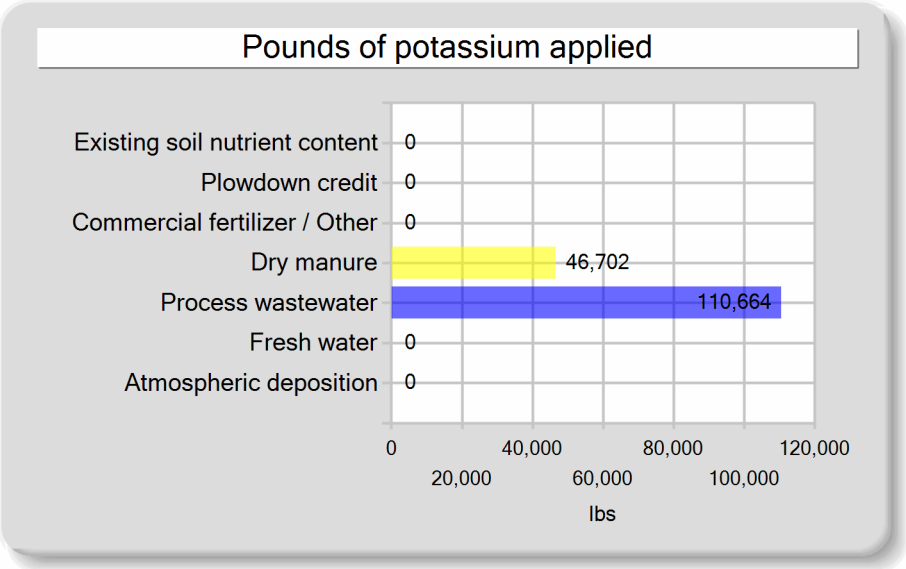
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

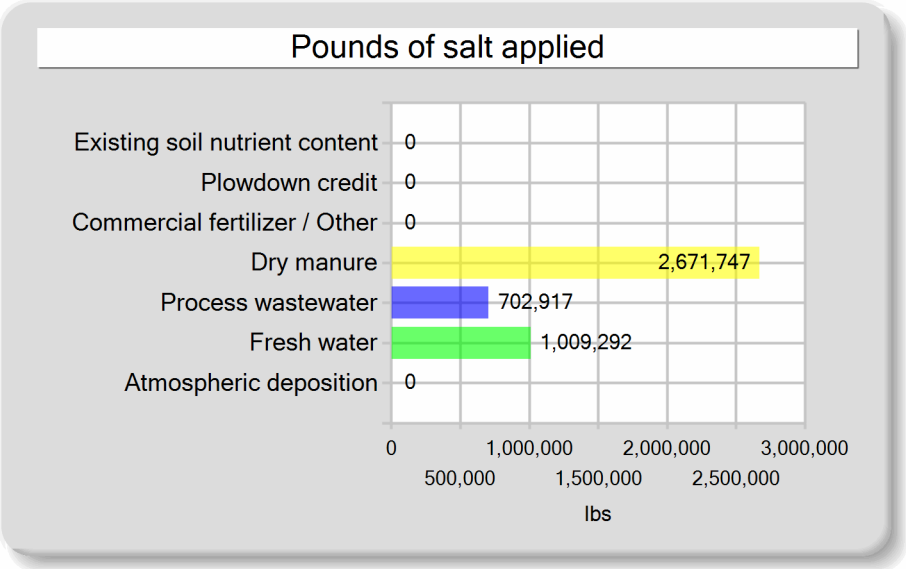


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Lower than anticipated nitrogen application rates (6.61-6.79 lbs of nitrogen removed vs 8 lbs of nitrogen anticipated) resulted in Fields F7 and R8 to have field ratios above 1.50. Applications were in line with targeted removal rates.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Rick Jones Rick Jones (May 21, 2024 18:12 PDT)	Eldred C. Pires Eldred Pires (May 20, 2024 17:53 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Rick & Julie Jones	Eldred & Amanda Pires
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 21, 2024	May 20, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
NW #1	T3303041A	Electrical Conductivity (EC)	EPA 120.1 and	2.02	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.648	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	970	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	6.25	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1290	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)			1.29
Electrical Conductivity (EC)	97.1			0.874 (T3303039B)	0.876 (Dup)	0.309
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)			2.29
Total Dissolved Solids (TDS)	99.0			500 (T3303041E)	490 (Dup)	2.02
Total Nitrogen (TN)	91.3			2.80 (T3297037A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 22.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
NW #2	T3303041B	Electrical Conductivity (EC)	EPA 120.1 and	2.01	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.603	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	975	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1290	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)			1.29
Electrical Conductivity (EC)	97.1			0.874 (T3303039B)	0.876 (Dup)	0.309
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)			2.29
Total Dissolved Solids (TDS)	99.0			500 (T3303041E)	490 (Dup)	2.02
Total Nitrogen (TN)	91.3			2.80 (T3297037A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 23.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Tower	T3303041C	Electrical Conductivity (EC)	EPA 120.1 and	2.01	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.792	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1030	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	3.59	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1290	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)			1.29
Electrical Conductivity (EC)	97.1			0.874 (T3303039B)	0.876 (Dup)	0.309
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)			2.29
Total Dissolved Solids (TDS)	99.0			500 (T3303041E)	490 (Dup)	2.02
Total Nitrogen (TN)	91.3			2.80 (T3297037A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 20.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
R & J	T3303041D	Electrical Conductivity (EC)	EPA 120.1 and	2.00	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.819	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	975	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1280	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)			1.29
Electrical Conductivity (EC)	97.1			0.874 (T3303039B)	0.876 (Dup)	0.309
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)			2.29
Total Dissolved Solids (TDS)	99.0			500 (T3303041E)	490 (Dup)	2.02
Total Nitrogen (TN)	91.3			2.80 (T3297037A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 21.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Alberto	T3303041E	Electrical Conductivity (EC)	EPA 120.1 and	1.03	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.921	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	500	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	3.72	mg/L
		Soluble Salts (SALT-SOL)	Calculation	660	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)			1.29
Electrical Conductivity (EC)	97.1			0.874 (T3303039B)	0.876 (Dup)	0.309
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)			2.29
Total Dissolved Solids (TDS)	99.0			500 (T3303041E)	490 (Dup)	2.02
Total Nitrogen (TN)	91.3			2.80 (T3297037A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Jones Farms
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F & R Ag Services
Sampled By: K.Jones
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	10/30	8:30	Irrigation Wells	NW #1	21.0						H7
2		8:45	Irrigation Wells	NW #2	22.0						H7
3		9:00	Irrigation Wells	Tower IW	23.4						H7
4		9:15	Irrigation Wells	R & J IW	20.8						H7
5		9:30	Irrigation Wells	Alberto IW	21.8						H7
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
1st		F & R Ag Services				
2nd		DENELE ANALYTICAL, INC.	10-30-23 2:35			T3303041

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Egghouse	T3303055A	Electrical Conductivity (EC)	EPA 120.1 and	2.02	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.682	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1290	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	97.1			0.309
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)	2.29

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Kristen	T3303055B	Electrical Conductivity (EC)	EPA 120.1 and	2.03	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.783	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1300	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	98.6			0.099
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)	2.29

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 22.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
R & J	T3303055C	Electrical Conductivity (EC)	EPA 120.1 and	2.00	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.781	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1280	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	98.6			0.099
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)	2.29

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Alberto	T3303055D	Electrical Conductivity (EC)	EPA 120.1 and	2.02	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.806	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1290	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	98.6			0.099
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)	2.29

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Shop	T3303055E	Electrical Conductivity (EC)	EPA 120.1 and	2.00	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.770	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1280	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	98.6			0.099
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)	2.29

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 24.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Sugs	T3303055F	Electrical Conductivity (EC)	EPA 120.1 and	3.53	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	24.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2260	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	98.6			0.099
Nitrate Nitrogen (NO3-N)	98.0			4.00

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Jones Farms
19133 W. 6th Ave.
Stevinson, CA 95374

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 23.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy	T3303055G	Electrical Conductivity (EC)	EPA 120.1 and	2.03	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.447	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1300	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	98.6			0.099
Nitrate Nitrogen (NO3-N)	98.0			4.00

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: F & R Ag Services

Client: Jones Farms
Mailing address:
(not facility address)
Phone#:

Sampled By: K.Jones
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	10/30	7:00	Domestic Wells	Egghouse	20.8		X				
2		7:15	Domestic Wells	Kristen	20.6		X				
3		7:30	Domestic Wells	R & J	22.6		X				
4		7:45	Domestic Wells	Alberto	21.0		X				
5		9:00	Domestic Wells	Shop	21.2		X				
6		9:15	Domestic Wells	Sugs	24.0		X				
7		9:30	Domestic Wells	Dairy	23.6		X				
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st		F & R Ag Services				
2nd		DENELE ANALYTICAL, INC.	10-30-23 2135			T3303055

Required to be submitted to any agency? (YES / NO)

2023 Annual Report-Pires Dairy Farms #2

Final Audit Report 2024-05-22

Created:	2024-05-20
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAbYi7GHYDw2zzIcvPK8Z10onilIXStlLR

"2023 Annual Report-Pires Dairy Farms #2" History

Document created by Kristen Jones (kjones@fragservices.com)
2024-05-20 - 7:38:39 PM GMT

Document emailed to Amanda S Pires (piresdairy@sbcglobal.net) for signature
2024-05-20 - 7:39:40 PM GMT

Email viewed by Amanda S Pires (piresdairy@sbcglobal.net)
2024-05-21 - 0:52:41 AM GMT

Signer Amanda S Pires (piresdairy@sbcglobal.net) entered name at signing as Eldred Pires
2024-05-21 - 0:53:55 AM GMT

Document e-signed by Eldred Pires (piresdairy@sbcglobal.net)
Signature Date: 2024-05-21 - 0:53:57 AM GMT - Time Source: server

Document emailed to mrsjulesj@fire2wire.com for signature
2024-05-21 - 0:53:59 AM GMT

Email viewed by mrsjulesj@fire2wire.com
2024-05-22 - 1:10:53 AM GMT

Signer mrsjulesj@fire2wire.com entered name at signing as Rick Jones
2024-05-22 - 1:12:47 AM GMT

Document e-signed by Rick Jones (mrsjulesj@fire2wire.com)
Signature Date: 2024-05-22 - 1:12:49 AM GMT - Time Source: server

Agreement completed.
2024-05-22 - 1:12:49 AM GMT